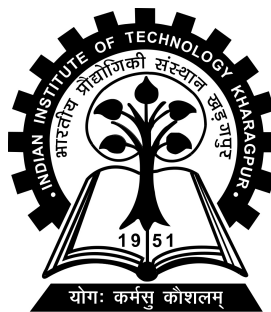


Combined Vector Field-Based Guidance and Control for UAVs

Project-I (AE47005) report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Dual Degree (B.Tech + M.Tech)
in
Aerospace Engineering

by
Dheeraj
(22AE30009)

Under the supervision of
Professor Sikha Hota



Department of Aerospace Engineering
Indian Institute of Technology Kharagpur

Autumn Semester, 2025-26

November 27, 2025

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

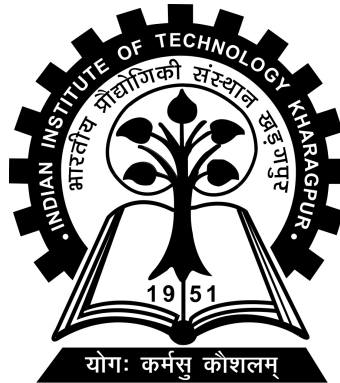
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CERTIFICATE

This is to certify that the project report entitled “Combined Vector Field-Based Guidance and Control for UAVs” submitted by Dheeraj (Roll No. 22AE30009) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Dual Degree (B.Tech + M.Tech) in Aerospace Engineering is a record of bona fide work carried out by him under my supervision and guidance during Autumn Semester, 2025-26.

Date: November 27, 2025
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Abstract

Name of the student: **Dheeraj**

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Degree for which submitted: **Dual Degree (B.Tech + M.Tech)**

Department: **Department of Aerospace Engineering**

Thesis title: **Combined Vector Field-Based Guidance and Control for UAVs**

Thesis supervisor: **Professor Sikha Hota**

Month and year of thesis submission: **November 27, 2025**

The thesis provides a comprehensive literature review of various path guidance algorithms, including both geometrical and non-geometrical approaches, as discussed in [1] and others. It focuses in detail on Vector Field (VF) based guidance for the autonomous navigation of Unmanned Aerial Vehicles (UAVs) in both 2D and 3D environments. The thesis also conducts the simulations based on the research conducted by Liang et al. in [2], which focuses on the kinematic aspects of UAV guidance using the Combined Vector Field (CVF) approach. This work extends their findings by validating the Combined Vector Field (CVF) approach with a comprehensive simulation of the quadrotor dynamics, incorporating Sliding Mode Controllers in MATLAB.

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Abbreviations

UAV	U nmanned A erial V ehical
MAV	M iniature A ir V ehicle
PI	P roportional I ntegral
PD	P roportional D erivative
PID	P roportional I ntegral D erivative
VTP	V irtual T arget P oint
LOS	L ine O f S ight
PLOS	P ure Pursuit and L ine O f S ight
NLGL	N on - L inear G uidance L aw
LQR	L inear Q uadratic R egulator
VF	V ector F ield
CVF	C ombined V ector F ield
VSC	V ariable S tructure C ontrol
SMC	S liding M ode C ontrol

Symbols

g	Gravitational Acceleration ($= 9.81 \text{ m/s}^2$)
ψ	Heading Angle
ψ_d	Desired Heading Angle
χ	Course Angle
χ_d	Desired Course Angle
γ	Climb Angle
γ_d	Desired Climb Angle
δ	Standoff Distance in Carrot Chasing Method
λ	Standoff Angle in Carrot Chasing Method

Chapter 1

Introduction

Unmanned Aerial Vehicles (UAVs) have become integral to modern aerospace applications, ranging from surveillance and reconnaissance to delivery and environmental monitoring. Ensuring precise and reliable navigation of UAVs is critical for their autonomous operation. Automatic guidance systems are designed to enable UAVs to follow desired trajectories or reach specified targets without human intervention. Over the years, several algorithms have been proposed to address the guidance problem, including nonlinear guidance laws, geometric methods, and vector field approaches [1].

Among these methods, Vector Field (VF) based guidance has gained substantial attention due to its robustness, simplicity, and ability to provide smooth convergence to desired trajectories. VF-based methods define a direction vector at every point in the space, enabling the UAV to follow paths while inherently correcting deviations. Early and influential contributions in VF-based path-following can be found in [3, 4], where VF guidance laws were successfully applied to miniature UAVs. Improvements addressing wind disturbances and environmental effects have also been demonstrated [5].

To further improve smoothness and flexibility, Combined Vector Field (CVF) approaches extend classical VF ideas by blending multiple vector fields to satisfy curvature and continuity constraints. Liang and Jia [2] developed a comprehensive CVF formulation that enables twice-differentiable path following in both 2-D and 3-D environments. Their approach serves as the foundation for the kinematic path-following simulations presented in this thesis. Related concepts combining geometric reasoning and VF-based strategies also appear in marine and ground vehicle applications [6, 7].

While kinematic-level guidance determines the desired direction of motion, the actual performance of the UAV depends critically on its dynamic behaviour. Quadrotors, being underactuated and nonlinear systems, require robust control strategies to ensure stable and precise trajectory tracking. Foundational modelling and control frameworks for quadrotors are provided in works such as [8, 9, 10, 11].

Sliding Mode Control (SMC) has emerged as a powerful nonlinear technique for robust UAV control due to its strong disturbance rejection properties. Early seminal work on SMC theory appears in [12], while practical engineering applications are explored in [13]. Specific implementations of SMC for quadrotors, demonstrating robust trajectory tracking and stability, can be found in [14, 15, 16]. These works show that SMC-based controllers can effectively handle parameter uncertainties, wind disturbances, and model inaccuracies, making them well-suited for dynamic UAV environments.

This thesis combines the CVF-based kinematic guidance of [2] with a full quadrotor dynamic model and an SMC-based controller [14, 15]. The objective is to evaluate how well VF-based guidance performs when integrated with a robust nonlinear controller. The simulation framework incorporates the dynamic behaviour of the quadrotor, enabling realistic assessment of convergence, tracking performance, and robustness.

The structure of this thesis is as follows. A detailed literature review of UAV guidance and control strategies is presented first, consolidating contributions such as [1, 3, 17, 18]. Next, the Combined Vector Field (CVF) approach is explained in depth, followed by the modelling of quadrotor dynamics and the implementation of the Sliding Mode Controller. Results from both kinematic and dynamic simulations are then presented and analysed. The thesis concludes with a discussion of limitations, contributions, and future research directions.

Chapter 2

Literature Review

Path following is a fundamental problem in UAV navigation, particularly crucial for ensuring autonomous and precise trajectory tracking. Over the years, significant research has been done to develop algorithms that enable UAVs to follow predefined 2-D paths accurately and robustly, despite environmental disturbances and vehicle constraints. This Chapter presents a comprehensive literature review of 2-D path-following approaches.

2.1 2-D Kinematics

The most commonly used paths for constructing more complex paths, as well as the design of path-following algorithms, are the line and circular (loiter) paths, assuming constant altitude and airspeed. The kinematics of any vehicle in a 2-dimensional x-y plane with constant altitude and airspeed assumptions can be given as,

$$\dot{x} = V_a \cos(\psi) + w_x = V_g \cos(\chi) \quad (2.1)$$

$$\dot{y} = v_a \sin(\psi) + w_y = V_g \sin(\chi) \quad (2.2)$$

and, assuming the first-order heading angle dynamics,

$$\dot{\psi} = k(\psi_d - \psi) \quad (2.3)$$

$$u = \text{max_limit}(\psi V_a) \quad (2.4)$$

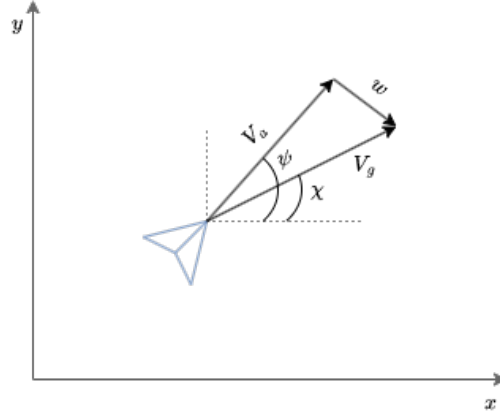


FIGURE 2.1: 2-D Kinematics

where,

ψ is the heading angle (angle with respect to the wind vector)

ψ_d is the desired heading angle

χ is the course angle (angle with respect to the ground x -axis)

V_a is the airspeed of the UAV (speed with respect to wind)

w_x is the wind component in the ground x -axis

w_y is the wind component in the ground y -axis

V_g is the resultant ground speed of the UAV

k is the proportional control gain of the heading controller

u is the commanded acceleration to the UAV (or commanded centripetal acceleration as the airspeed is constant, and only the direction of the velocity changes)

2.2 Path Following Methods

There are many path-following methods, which can be broadly classified into two categories: *Geometric* and *Non-Geometric* path-following methods.

2.2.1 Geometric Methods

There are many path-following methods that employ a purely geometric approach to reach the desired path. Out of which **Carrot-Chasing**, **Non-Linear Guidance Law (NLGL)**, and **Pure Pursuit and Line of Sight (LOS) Based Path Following (PLOS)** methods are discussed in this section.

NOTE: In this section, the methods only discuss *Line* and **Loiter** paths as they are often used to create complex paths. The discussion also doesn't include the presence of wind in these methods.

2.2.1.1 Carrot - Chasing Method

This method utilizes the concept of the Virtual Target Point (VTP), which is essentially the point on the desired path that the UAV attempts to follow. This VTP on the path is generally called the "carrot", and hence the name "Carrot-Chasing" or "Rabbit-Chasing". Now, as the Rabbit/UAV comes closer to the VTP, a specified standoff distance (δ) is maintained between the UAV and the VTP at all times. This standoff distance (δ) is the path parameter, which governs the convergence and stability of the UAV to the path (Line or Circle).

Line Path Following Following the line path with the Carrot-Chasing method is straightforward.

- Initialize the line to follow by specifying two points on the line. (Generally, specified by two distinct points in the line)

- Initialize all the parameters such as Initial Position $p_0(x_0, y_0)$, Initial Heading angle (ψ), constant UAV airspeed (V_a), path parameter (δ), proportional controller gain (k), and minimum turn radius (R_{min}) to impose the *Curvature Constraint*.

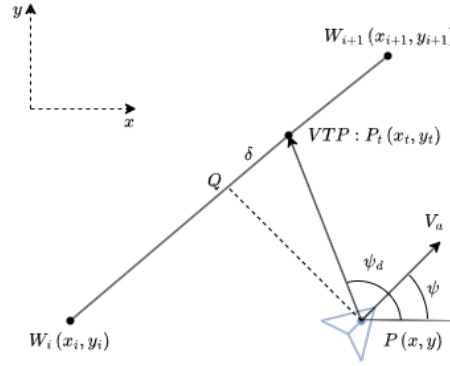


FIGURE 2.2: Schematic of the implementation of the 2-D Carrot-Chasing method for line path following

- Compute the perpendicular projection of the UAV position onto the line, and compute the position of the VTP by incrementing the standoff distance to the actual projection of the UAV on the line.
- Compute the desired heading angle (ψ_d), which directs to the VTP.

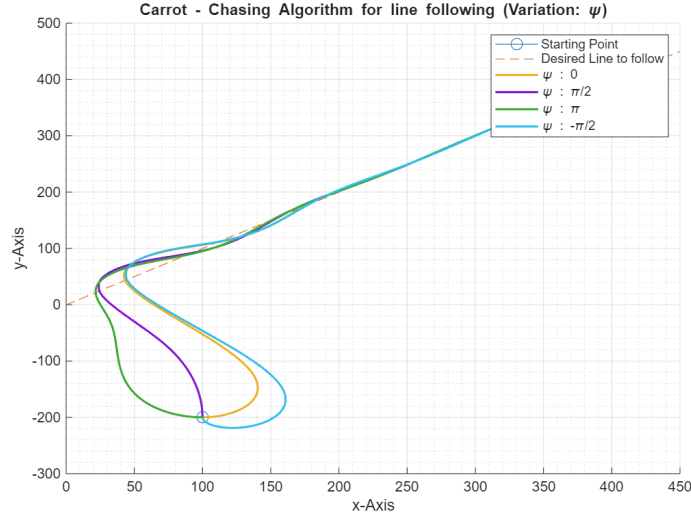
$$\psi_d = \text{atan2}(y_d, x_d), \psi_d \in (-\pi, \pi] \quad (2.5)$$

- Compute the new heading angle rate using the control law,

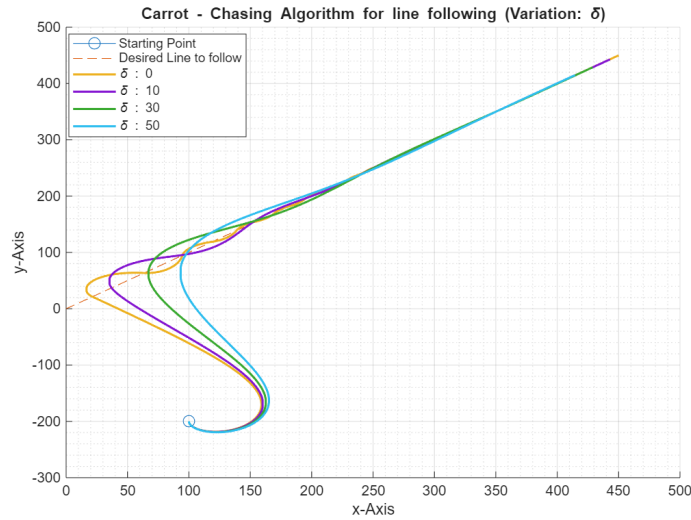
$$\dot{\psi} = k(\psi_d - \psi) \quad (2.6)$$

- Saturate the calculated rate with maximum allowable turn rate $\dot{\psi}_{max} = \frac{V_a}{R_{min}}$
- Now, calculate the new Heading angle (ψ) using Euler integration or any other integration methods, and repeat all the steps from step 3.

Simulation For the Simulation of this method for Line Path Following, the line to follow is given by these two points $(0, 0) m$ and $(450, 450) m$ with initial position $P_0 (100, 0) m$, constant airspeed $25m/s$, minimum turn radius $10m$, proportional controller gain 0.5 and the initial heading angle (ψ_0) and standoff distance (δ) are varied.



(a) Variation in Initial Heading angle (ψ_0) with $\delta = 15$



(b) Variation in Standoff distance (δ) with $\psi_0 = -\frac{\pi}{2}$

FIGURE 2.3: Line Path Following using Carrot Chasing Method

Loiter Path Following Loiter Path following using the Carrot Chasing Method is similar to the line path following. The main difference is that the standoff distance (δ) specified in the case of line path following is replaced with standoff angle (λ) in loiter path following.

- Specify the loiter path to follow by defining the center $O(x_o, y_o)$, and the radius r of the circle.
- Initialize the parameters as in Line path following (Step 2).

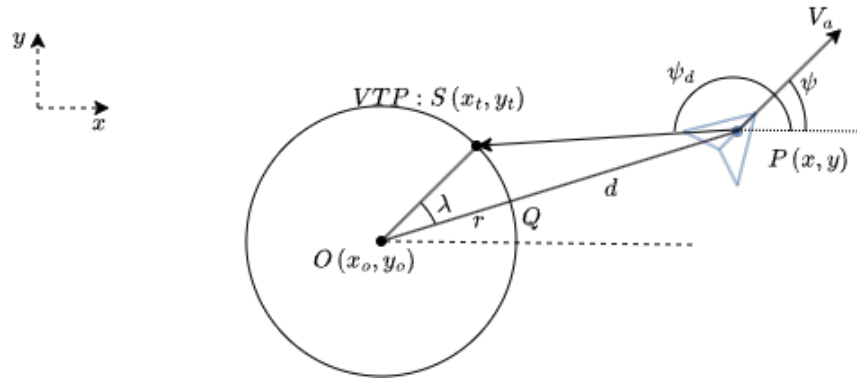
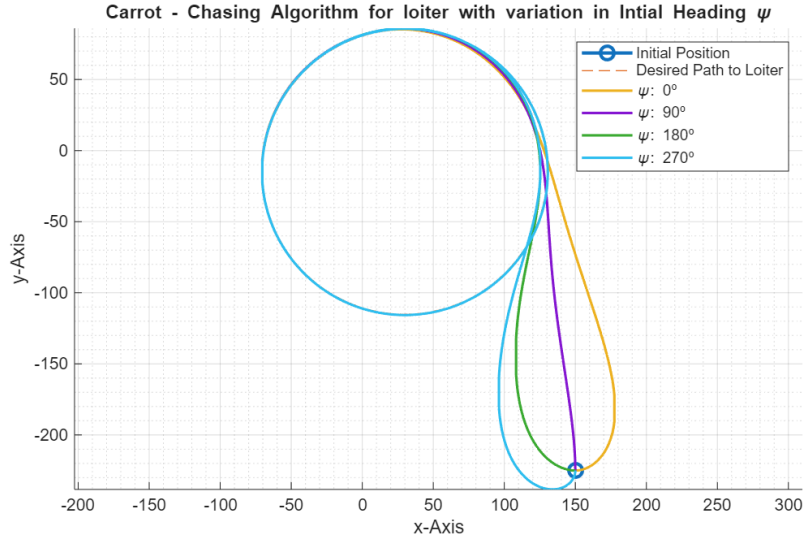


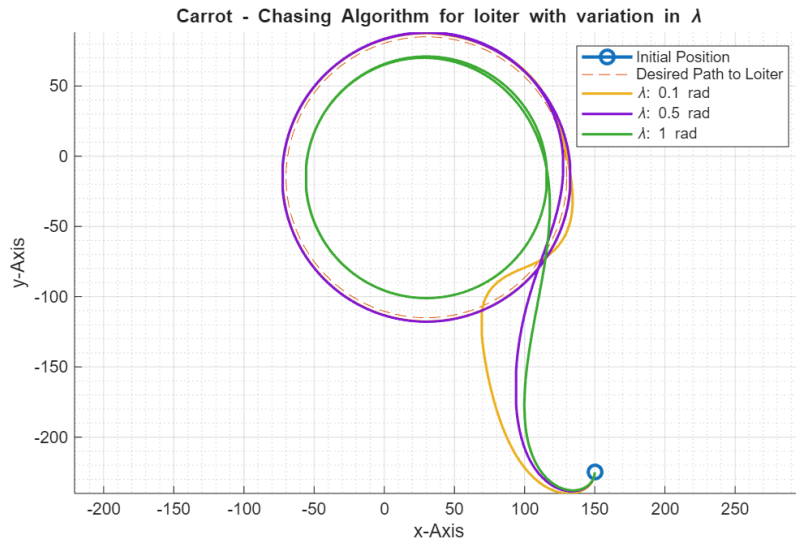
FIGURE 2.4: Schematic of the implementation of the 2-D Carrot-Chasing method for loiter path following

- Calculate the angle between the +ve x-axis and the line joining the loiter circle center and the current UAV position.
- Now, add/subtract the specified standoff angle from the previously calculated in the previous step (Add the standoff angle for clockwise loiter and subtract for anticlockwise).
- Calculate the new VTP using the angle obtained in the previous step.
- Now, again do similar as in line path following, calculate desired heading angle, calculate heading angle rate, saturate it, and find the updated heading angle, and repeat from step 3 for the next time step.

Simulation Similarly, for the Simulation of Carrot Chasing method for Loiter Path Following, the Circle to follow is given by the center $(30, -15) m$ and the radius $100m$ with initial position $P_0(150, -225) m$, constant airspeed $25m/s$, minimum turn radius $10m$, proportional controller gain 0.8 and the initial heading angle (ψ_0) and standoff angle (λ) are varied.



(a) Variation in Initial Heading angle (ψ_0) with $\lambda = 0.6$



(b) Variation in Standoff angle (λ) with $\psi_0 = -\frac{\pi}{2}$

FIGURE 2.5: Loiter Path Following using Carrot Chasing Method

2.2.1.2 Non-Linear Guidance Law (NLGL)

The Non-Linear Guidance Law approach also incorporates the concept of a Virtual Target Point (VTP), but it determines the VTP differently than the Carrot Chasing method. Additionally, instead of using a simple Proportional-Derivative (PD) controller to point the UAV to the desired heading, as in the Carrot Chasing approach, it employs a Non-Linear Guidance Law, as its name implies.

Line Path Following For the Line Path Following using the Non-Linear Guidance Law, we first have to have a method to determine the VTP to follow. To determine the VTP over the line to follow, we draw a finding circle of radius L (predefined) from the current position of the UAV and find the intersection points between the finding circle and the line. If the finding circle and the line to follow intersect, then there must be at least one intersection point.

If there are two intersection points, then one of the intersection points, with a higher or lower value, is consistently chosen depending on the direction of the line to follow. In case there is only one intersection point, this implies that the line to follow is tangent to the finding circle, and the only intersection point is our VTP.

Now, if the finding circle radius is small or the UAV is positioned very far away from the line to follow, then there will be no intersection points. In that case, the intersection point between the line perpendicular to the line to follow passing through the current UAV position and the line to follow will be the VTP.

- Initialize the line to follow by specifying two points on the line (w_i and w_{i+1}).
- Initialize all the parameters such as Initial Position $p_0 (x_0, y_0)$, Initial Heading Angle (ψ), constant UAV airspeed (V_a), Radius of the finding circle (L), and minimum turn radius (R_{min}) to impose the *Curvature Constraint*.
- Compute the VTP to follow as discussed earlier.
- Compute the desired heading angle (ψ_d), which directs to the VTP.

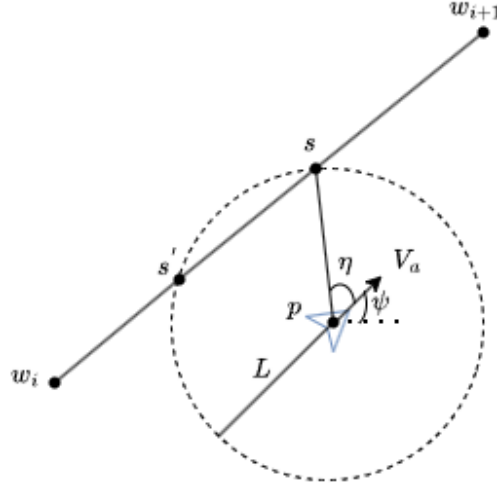


FIGURE 2.6: Schematic of finding the VTP for 2-D Line Path Following using NLGL

- Compute the new heading angle rate using the Non-linear control law,

$$\dot{\psi} = \frac{2V_a \sin(\eta)}{L} \quad (2.7)$$

where, $\eta = \psi_d - \psi$

- Saturate the calculated rate with the maximum allowable heading rate, $\dot{\psi}_{max} = \frac{V_a}{R_{min}}$
- Now, calculate the new heading angle (ψ) by integrating. Then, repeat all the steps from step 3 again.

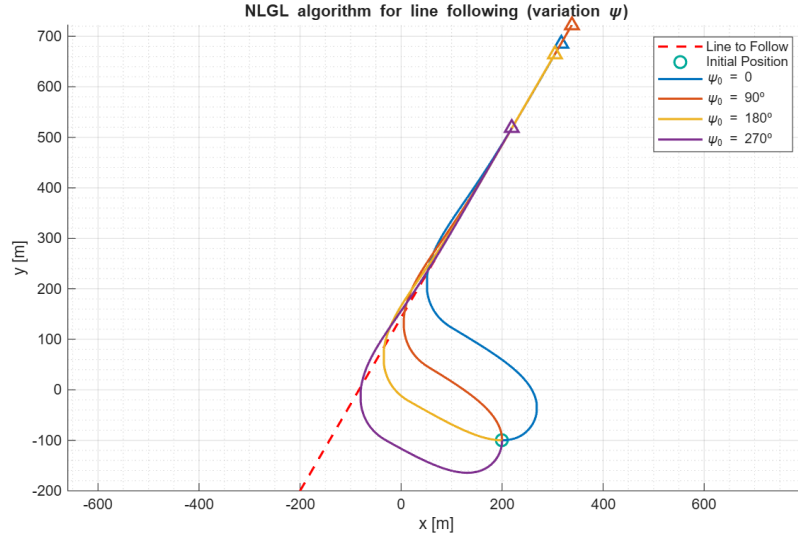
Simulation For the Simulation of this method for Line Path Following, the line to follow is given by these two points $(-200, -200) m$ and $(150, 400) m$ with initial position $P_0(200, -100) m$, constant airspeed $25 m/s$, maximum allowable roll angle ($\phi_{max} = 45^\circ$), and the initial heading angle (ψ_0) and radius of the finding circle (L) are varied.

Now, as the maximum heading rate $\dot{\psi}_{max}$ can be given as,

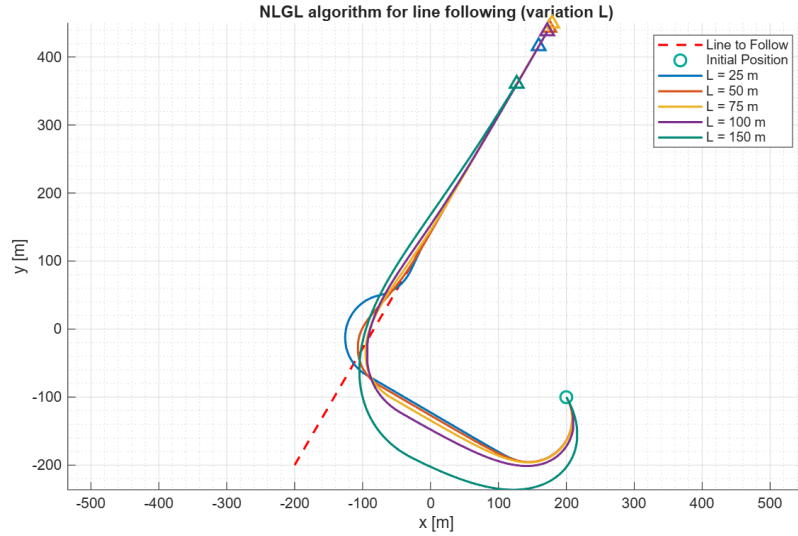
$$\dot{\psi}_{max} = \frac{g \tan(\phi_{max})}{V_a} \quad (2.8)$$

This implies,

$$\dot{\psi}_{max} = 0.3924 \text{ rad/s}^2$$



(a) Variation in Initial Heading angle (ψ_0) with $L = 100 \text{ m}$



(b) Variation in radius of finding circle (L) with $\psi_0 = -\frac{\pi}{3}$

FIGURE 2.7: Line Path Following using Non-Linear Guidance Law

Loiter Path Following Similar to Line Path Following, for Loiter Path Following, the intersection points are calculated between the circle to follow and the finding circle. Again, if there is only one intersection point, then that intersection point becomes the VTP. If there are two intersection points, then again, a higher

or lower value is consistently selected for the VTP, according to the direction of the loiter. Now, again, if there are no intersection points, then the VTP is chosen as the intersection point between the line joining the current UAV position and the center of the loiter circle, with the loiter circle.

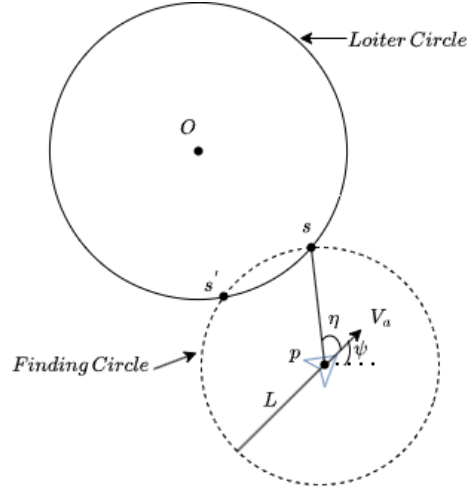
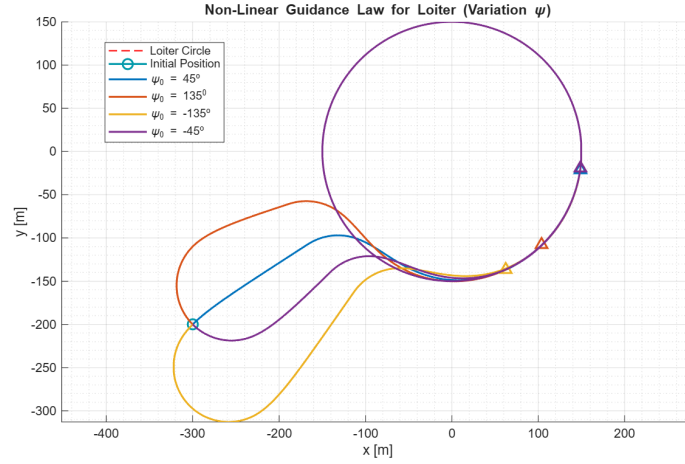


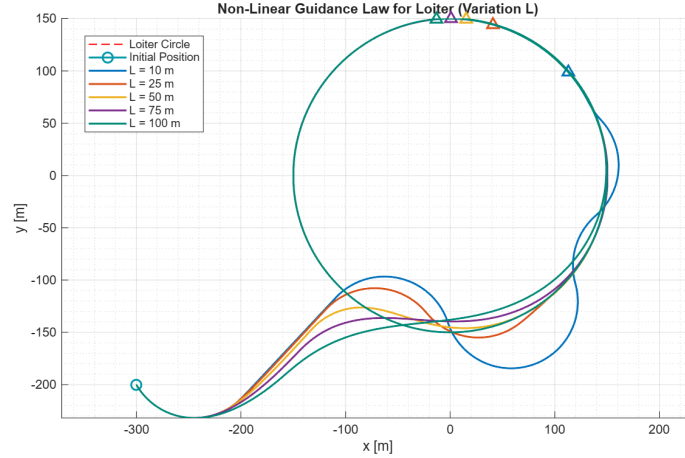
FIGURE 2.8: Schematic of finding the VTP for 2-D Loiter Path Following using NLGL

- Specify the loiter path to follow by defining the center $O(x_0, y_0)$, and the radius r of the circle.
- Initialize all the parameters as in Line Path Following using NLGL. (Step 2)
- Calculate the VTP as discussed earlier for the Loiter paths, and hence calculate the desired heading angle (ψ_d).
- Compute the new heading angle rate using the Non-Linear Control Law, as described in (2.7).
- Again, saturate the calculated rate with the maximum allowable heading rate.
- Now, calculate the new heading angle (ψ) by integrating. Then, repeat all the steps from step 3 for each time step.

Simulation Similarly, for the Simulation of Non-Linear Guidance Law for Loiter Path Following, the Circle to follow is given by the center $(0, 0) m$ and the radius $150 m$ with initial position $P_0 (-300, -200) m$, constant airspeed $25 m/s$, maximum heading rate $0.3924 rad/s^2$ and the initial heading angle (ψ_0) and the radius of the finding circle (L) are varied.



(a) Variation in Initial Heading angle (ψ_0) with $L = 50 m$



(b) Variation in radius of finding circle (L) with $\psi_0 = -\frac{\pi}{3}$

FIGURE 2.9: Loiter Path Following using Non-Linear Guidance Law

2.2.1.3 Pure Pursuit and Line of Sight (PLOS) based Path Following

Pure Pursuit and Line of Sight (PLOS) path following, as the name suggests, is the combination of Pure Pursuit-based path following and LOS-based path following. The Pure Pursuit part of it wants to converge it to a point, while the LOS part steers it to be aligned to the path itself. The combination of these two guidance laws with different gains will cause the UAV to follow the desired path. The guidance law for the Pure Pursuit is

$$\psi_p = k_1 (\theta_d - \psi)$$

where k_1 is the gain and θ_d is the desired angle (desired pure pursuit angle). The LOS guidance law ensures that the UAV follows the path to the pursuit point. The LOS guidance law is given by

$$\psi_l = \tan^{-1}(k_2 d), \quad \psi_l \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

where k_2 is the gain and d is the cross-track error from the path to follow. The Combined PLOS guidance law can be given as,

$$\dot{\psi} = k_1 (\theta_d - \psi) + \tan^{-1}(k_2 d) \quad (2.9)$$

where ψ_d is the desired heading angle, obtained from the PLOS guidance law.

Line Path Following For Line Path Following using PLOS guidance law, is straightforward:

- Initialize the line to follow by specifying two points on the line (w_i and w_{i+1}).
- Initialize all the parameters such as Initial Position $p_0(x_0, y_0)$, Initial Heading Angle (ψ), constant UAV airspeed (V_a) and minimum turn radius (R_{min}) to impose the *Curvature Constraint*.

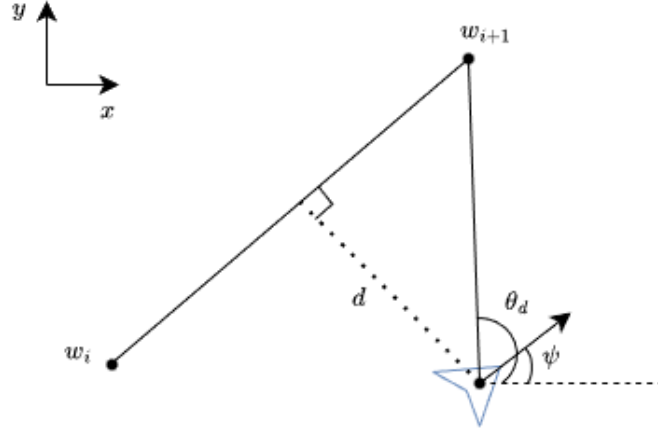
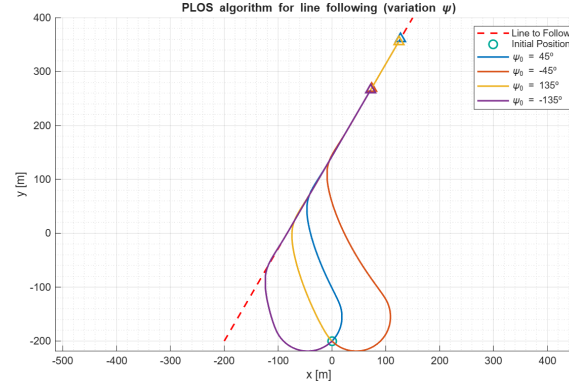


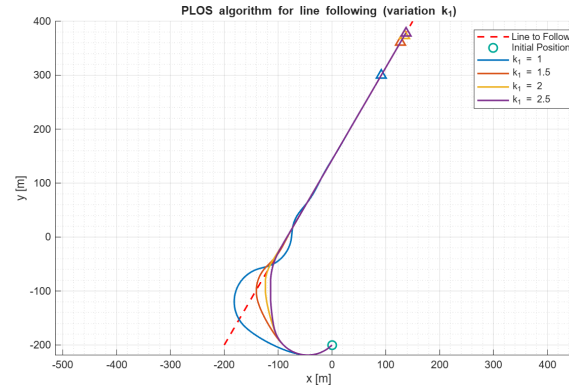
FIGURE 2.10: Schematic of finding the desired Pursuit angle (θ_d) and cross-track error (d) for 2-D Line Path Following using PLOS

- Compute the angle between $+x$ -axis and the line joining the current UAV position and the pursuit point (in this case w_{i+1}). This is the desired pursuit angle θ_d .
- Also compute the perpendicular distance between the current UAV position from the line to follow. This is the cross-track error d .
- Now, compute the heading angle rate ($\dot{\psi}$) using the combined PLOS guidance law in (2.9).
- Saturate the heading rate with the specified heading rate constraint.
- Integrate the heading angle for the next time step using the heading angle rate calculated in the step above.
- Repeat all the steps from step 3 for the next time steps.

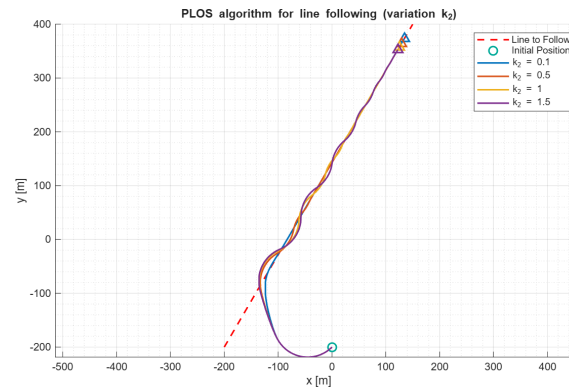
Simulation For the Simulation of Pure Pursuit and Line of Sight based guidance law for Line Path Following, the Line to follow is given by the two points $w_1 (-200, -200) m$ and $w_2 (150, 400) m$ with initial position $P_0 (0, -200) m$, constant airspeed $25m/s$, maximum allowable roll 45° and initial heading angle (ψ_0), pursuit guidance gain (k_1) and LOS guidance gain (k_2) are varied subsequently.



(a) Variation in Initial Heading angle (ψ_0) with gains $k_1 = 2$ and $k_2 = 0.1$



(b) Variation in Pursuit guidance gain (k_1) with $\psi_0 = -\frac{3\pi}{4}$, and $k_2 = 0.1$



(c) Variation in LOS guidance gain (k_2) with $\psi_0 = -\frac{3\pi}{4}$, and $k_1 = 2$

FIGURE 2.11: Line Path Following using Pure Pursuit and Line of Sight (PLOS) based guidance

Loiter Path Following Similar to Line Path Following,

- Initialize the loiter circle to follow by specifying the center $O(x_o, y_o)$ and the radius r .
- Initialize all the parameters as in PLOS line path following step 2.

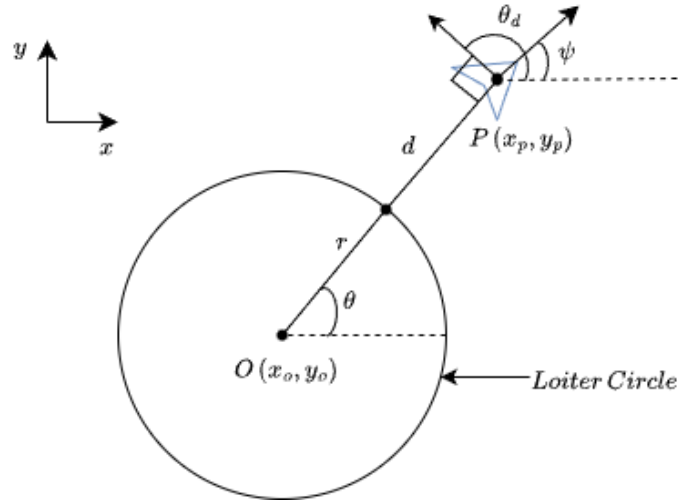


FIGURE 2.12: Schematic of finding the desired Pursuit angle (θ_d) and cross-track error (d) for 2-D Loiter Path Following using PLOS

- Calculate the desired Pursuit angle (θ_d) and cross-track error (d), as shown in figure 2.12.

$$\theta_d = \tan^{-1} \left(\frac{y_p - y_o}{x_p - x_o} \right) + s \frac{\pi}{2}, \quad \theta_d \in (-\pi, \pi] \quad (2.10)$$

$$d = ||P - O|| - r \quad (2.11)$$

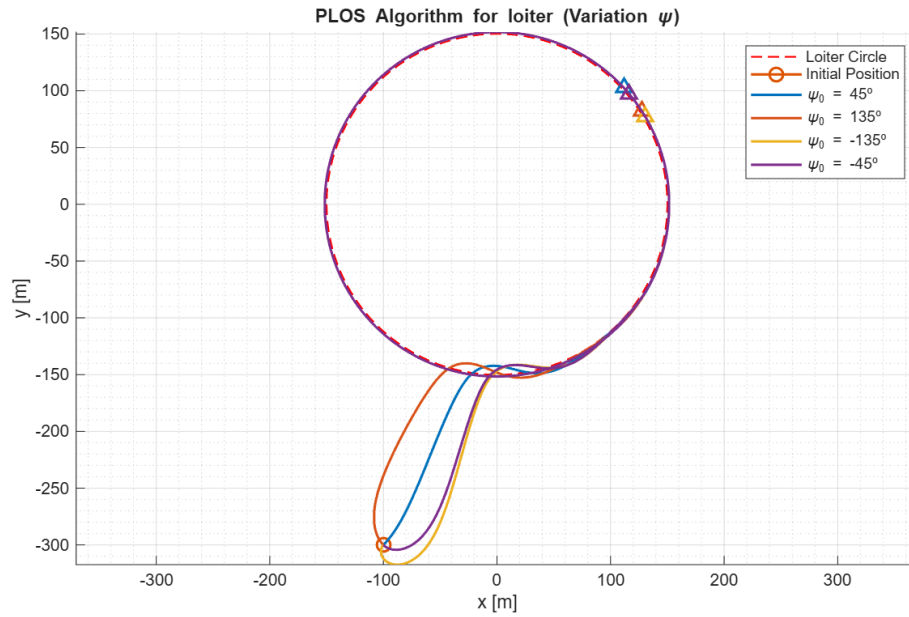
where s can be $+1$ for anticlockwise loitering and -1 for clockwise loitering.

- Now, compute the heading angle rate ($\dot{\psi}$) using the combined PLOS guidance law,

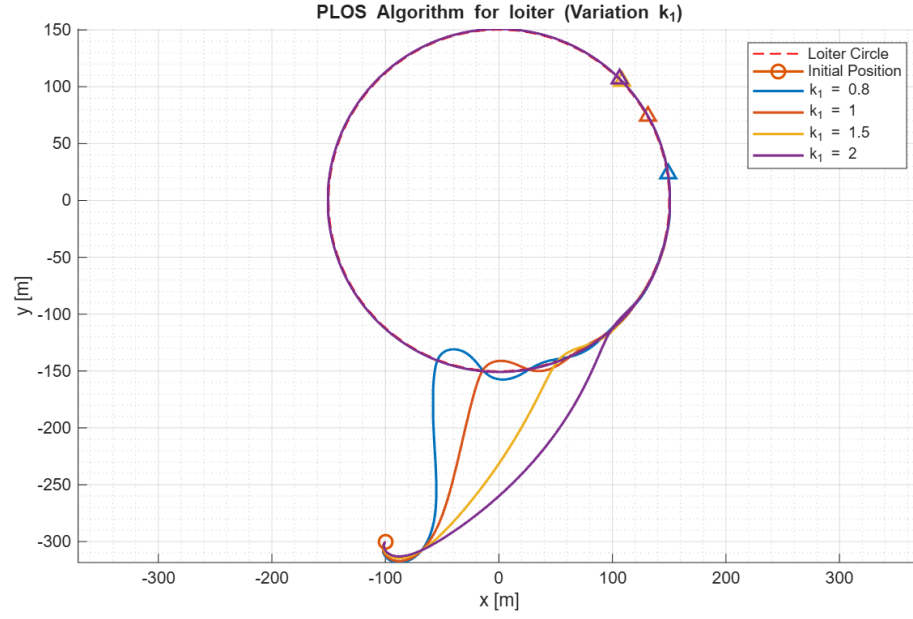
$$\dot{\psi} = k_1 (\theta_d - \psi) + s \tan^{-1} (k_2 d) \quad (2.12)$$

- Saturate the heading rate with the specified heading rate constraint.
- Integrate the heading angle for the next time step using the heading angle rate calculated in the step above.
- Repeat all the steps from step 3 for the next time steps.

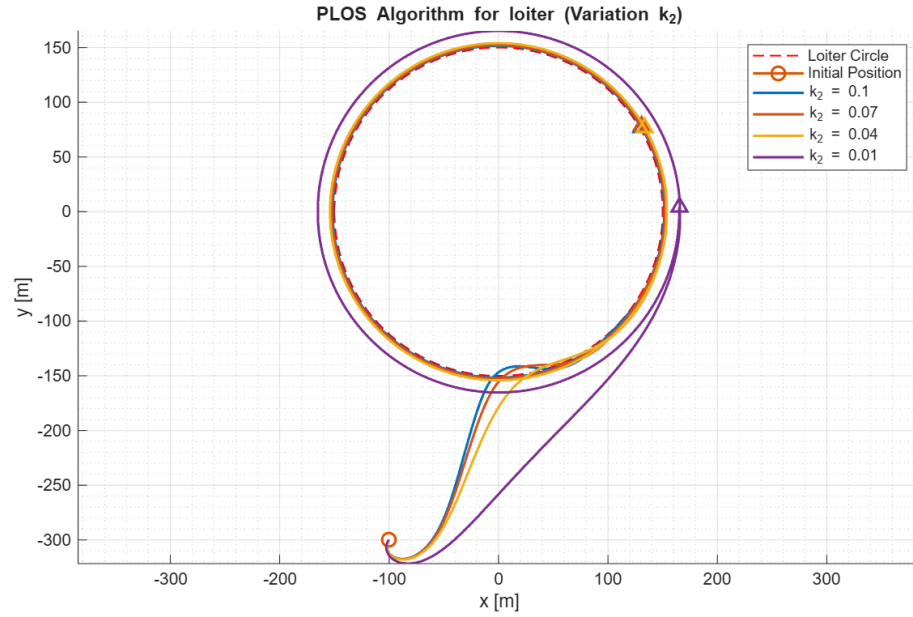
Simulation Similarly, for the Simulation of Pure Pursuit and LOS guidance for Loiter Path Following, the Circle to follow is given by the center $(0,0) m$, the radius $150 m$ and anticlockwise ($s = 1$) direction to loiter with initial position $P_0(-100, -300) m$, constant airspeed $25 m/s$, maximum allowable roll angle 45° and initial heading angle (ψ_0), pursuit guidance gain (k_1) and LOS guidance gain (k_2) are varied subsequently.



(a) Variation in Initial Heading angle (ψ_0) with gains $k_1 = 1$ and $k_2 = 0.1$



(b) Variation in Pursuit guidance gain (k_1) with $\psi_0 = -\frac{3\pi}{4}$, and $k_2 = 0.2$



(c) Variation in LOS guidance gain (k_2) with $\psi_0 = -\frac{3\pi}{4}$, and $k_1 = 1$

FIGURE 2.13: Loiter Path Following using Pure Pursuit and Line of Sight (PLOS) based guidance

2.2.2 Non-Geometric Methods

There are many non-geometric methods, such as **Vector Field (VF)** based, **Linear Quadratic Regulator (LQR)** based, etc. However, the discussion in this thesis primarily focuses on the **Vector Field-based Path Following**.

2.2.2.1 Vector Field (VF) based Path Following

Vector field-based path-following methods generate a spatial vector field that denotes the desired direction of travel for the UAV. These Vector fields can be formulated, and the argument of stability can also be provided using **Lyapunov Stability Proof**. Compared to the "Way Point" path following methods, VF-based path following methods generally exhibit reduced cross-track oscillations and can also handle wind gracefully.

The Vector fields for following the line and loiter paths are given in figures 2.14(a) and 2.14(b).

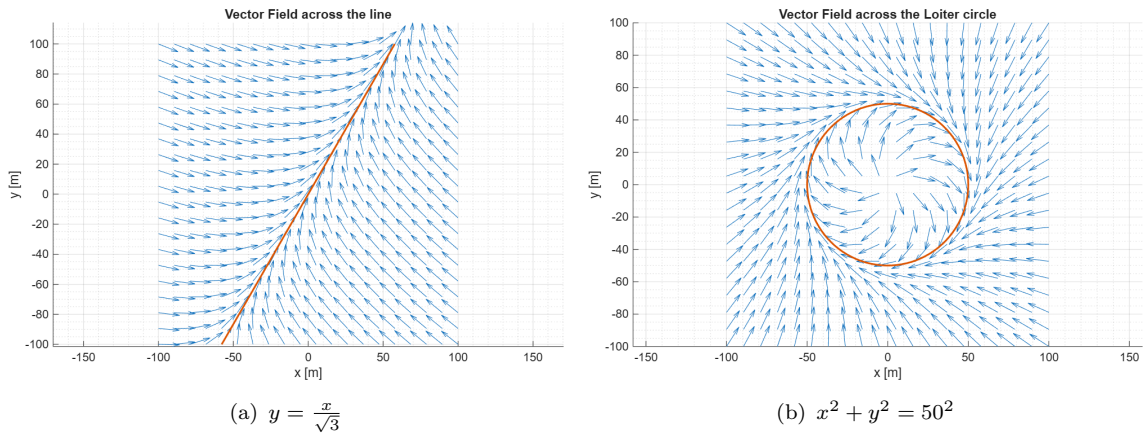


FIGURE 2.14: Vector Fields for line and loiter path following

Line Path Following For straight line Path following, a desired course vector field is constructed. Let d be the lateral distance of the UAV from the path and χ be the difference between the direction of the path and the course of the UAV. Since our objective is to construct the vector field so that, when d is large, the UAV is directed to approach the path with course angle $\chi^\infty(0, \pi/2)$, and that as d approaches zero, the course also approaches zero. Define the desired course of the UAV as,

$$\chi^d(d) = -\chi^\infty \frac{2}{\pi} \tan^{-1}(kd) \quad (2.13)$$

where k is a positive constant that influences the convergence of the course angle from χ^∞ to 0. Large values of k result in short, abrupt transitions, while small values of k cause long, smooth transitions to the desired course. The proof for this Vector Field to make the UAV go asymptotically to the desired path can be found in [4].

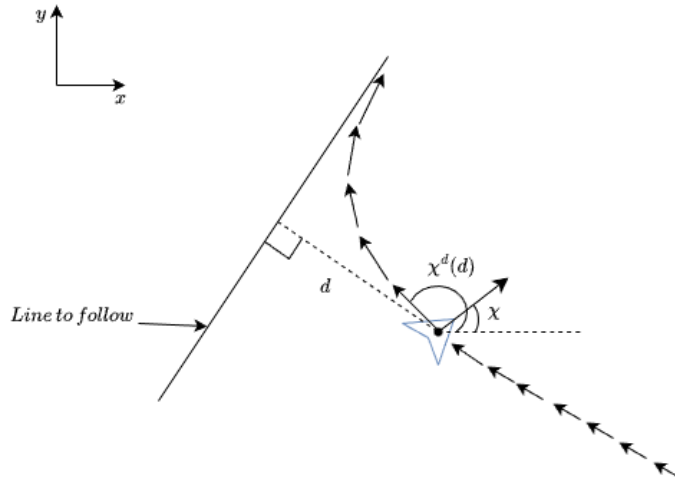


FIGURE 2.15: Schematic of the line path following using VF-based guidance

Now, to make the course angle to the desired course angle in finite time, a *Sliding Mode Controller* with a sliding surface as

$$S = \{(d, \chi) : \chi = \chi^d(d)\} \quad (2.14)$$

Let the error in the course angle,

$$\tilde{\chi} = \chi - \chi^d(d) \quad (2.15)$$

upon differentiating with time,

$$\dot{\tilde{\chi}} = \dot{\chi} - \dot{\chi}^d(d) \quad (2.16)$$

Assuming the UAV has a first-order course hold in the outer loop, such that $\dot{\chi} = \alpha(\chi^c - \chi)$, where χ^c is the commanded course angle to the outer loop, then substituting in (2.16).

$$\dot{\tilde{\chi}} = \alpha(\chi^c - \chi) + \chi^\infty \frac{2}{\pi} \frac{k}{1 + (kd)^2} V_g \sin \chi \quad (2.17)$$

Let the Lyapunov candidate $W = (1/2) \tilde{\chi}^2$ and take the derivative to obtain

$$\begin{aligned} \dot{W} &= \tilde{\chi} \dot{\tilde{\chi}} \\ &= \tilde{\chi} (\dot{\chi} - \dot{\chi}^d(d)) \\ &= \tilde{\chi} \left(\alpha(\chi^c - \chi) + \chi^\infty \frac{2}{\pi} \frac{k}{1 + (kd)^2} V_g \sin \chi \right) \end{aligned}$$

If we choose the $\dot{\tilde{\chi}} = -\kappa \text{sign}(\tilde{\chi})$, then the above derivative to the Lyapunov candidate becomes less than zero $\dot{W} = -\kappa \tilde{\chi} \text{sign}(\tilde{\chi})$. This in turn implies that the course error $\tilde{\chi}$ goes to zero in an exponential manner. Therefore,

$$\chi^c = \chi - \frac{1}{\alpha} \chi^\infty \frac{2}{\pi} \frac{k}{1 + (kd)^2} V_g \sin \chi - \frac{\kappa}{\alpha} \text{sign}(\tilde{\chi}) \quad (2.18)$$

where

$$\text{sign}(x) = \begin{cases} 1, & \text{if } x > 0 \\ 0, & \text{if } x = 0 \\ -1, & \text{if } x < 0 \end{cases}$$

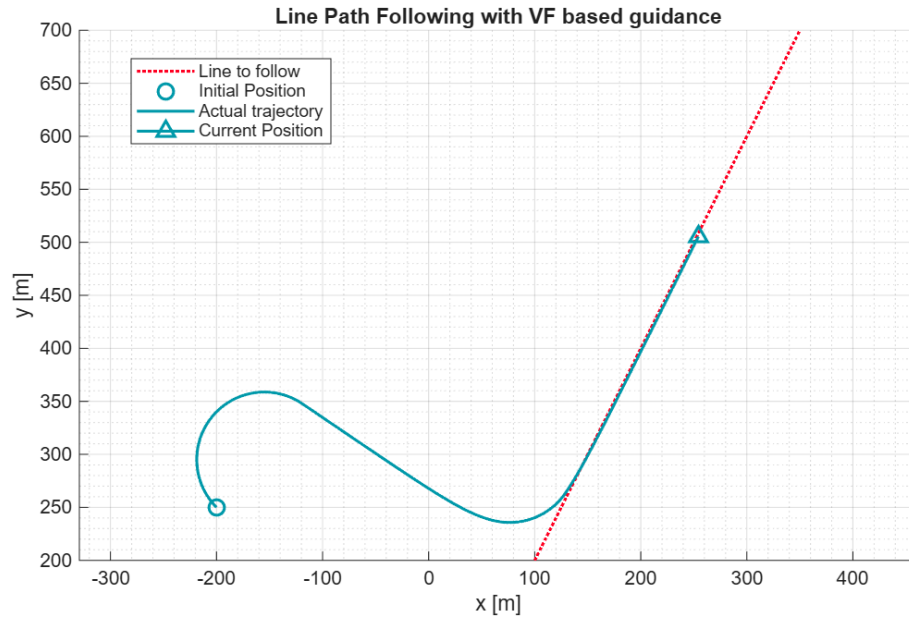
It is well known that the sign function leads to chattering in the control. To mitigate the effects of chattering, the sign function is replaced with a saturation function with a boundary region across the sliding surface of width ϵ .

$$\chi^c = \chi - \frac{1}{\alpha} \chi^\infty \frac{2}{\pi} \frac{k}{1 + (kd)^2} V_g \sin \chi - \frac{\kappa}{\alpha} \text{sat} \left(\frac{\tilde{\chi}}{\epsilon} \right) \quad (2.19)$$

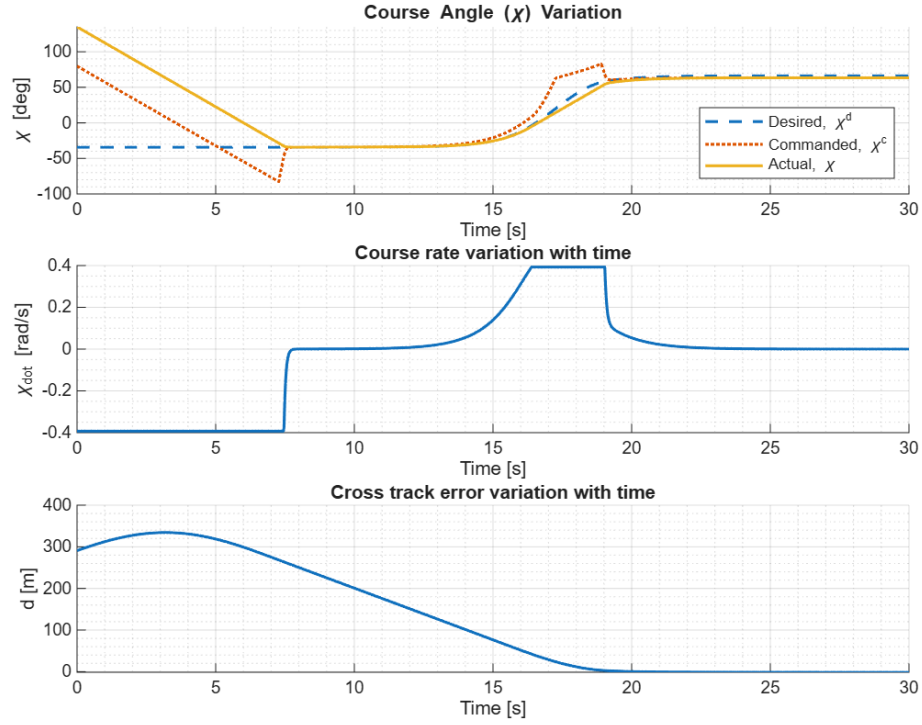
where,

$$\text{sat}(x) = \begin{cases} x, & \text{if } |x| < 1 \\ \text{sign}(x), & \text{otherwise} \end{cases}$$

Simulation For Simulation of the line path following using the VF-based guidance, the line to follow is given as the starting point $(100, 200) m$, with the direction vector $[0.5, 1]$ and length $500m$. The UAV parameters such as constant airspeed $25 m/s$, initial position $(-200, 200) m$, initial course angle $3\pi/4 rad$, course angle at infinity $\chi^\infty = \pi/2 - \text{atan2}(\text{direc}(2), \text{direc}(1)) = 2.779 rad$, course hold controller gain $(\alpha =) 1.65$, SMC gain $(\kappa =) \pi/2$ and width of the boundary region $(\epsilon =) 0.1$ and the maximum allowable roll angle is 45° .



(a) Trajectory



(b) Different parameters Variations with time while line path following with VF guidance

FIGURE 2.16: Line Path Following with VF-based guidance

Loiter Path Following The Algorithm for circular orbits creates vector fields in a manner similar to the straight lines. Consider the desired orbit path shown in figure 2.17. In this discussion, an anticlockwise orbit will be considered. The development for the clockwise orbits is similar, but several sign changes will be required. Let the center of the orbit be $O(x_o, y_o)$ and the radius be r . When the distance between the UAV and the center of the orbit is large, it is desirable for the UAV to fly towards the center of the orbit. If we define d as the radial distance of the UAV from the center of the orbit, then, when d is significantly larger than r , the desired course is,

$$\chi^d \approx \gamma - \pi$$

where γ is defined as the angular position of the UAV from the orbit center.

when $d = r$, the desired course angle $\chi^d = \gamma - \pi/2$ (+1 as anticlockwise orbit).

Therefore, the desired course angle can be given as,

$$\chi^d(d) = \gamma - \frac{\pi}{2} - \tan^{-1}(k(d - r)) \quad (2.20)$$

The stability proof of this algorithm is given using Lyapunov stability proof in [4].

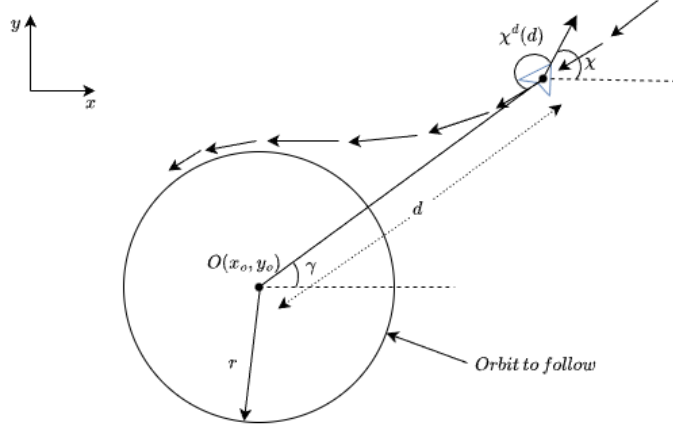


FIGURE 2.17: Schematic of the loiter path following using VF-based guidance

For orbit following, consider navigational dynamics in polar coordinates in terms of d and γ . Then, $x = x_o + d \cos(\gamma)$ and $y = y_o + d \sin(\gamma)$, where (x_o, y_o) is the center of the orbit. Now, taking the derivative and simplifying both equations,

$$\dot{d} = V_g \cos(\chi - \gamma) \quad (2.21)$$

$$\dot{\gamma} = \frac{V_g}{d} \sin(\chi - \gamma) \quad (2.22)$$

Again, considering first-order course hold dynamics is given as,

$$\dot{\chi} = \alpha(\chi^c - \chi) \quad (2.23)$$

Using the same sliding mode approach as in Straight line path following, the sliding surface is given as

$$S = \{(d, \chi) : \chi = \chi^d(d)\} \quad (2.24)$$

Define $\tilde{\chi} = \chi - \chi^d(d)$ and differentiate to obtain,

$$\begin{aligned}\dot{\tilde{\chi}} &= \alpha(\chi^c - \chi) - \dot{\chi}^d(d) \\ &= \alpha(\chi^c - \chi) - \dot{\gamma} + \frac{k}{1 + (k(d - r))^2} \dot{d}\end{aligned}$$

substituting $\dot{d}$ and $\dot{\gamma}$ and also defining $\beta = k/(1 + (k(d - r))^2)$,

$$\dot{\tilde{\chi}} = \alpha(\chi^c - \chi) - \frac{V_g}{d} \sin(\chi - \gamma) + \beta V_g \cos(\chi - \gamma) \quad (2.25)$$

Letting the Lyapunov candidate $W = (1/2)\tilde{\chi}^2$, and taking the derivative

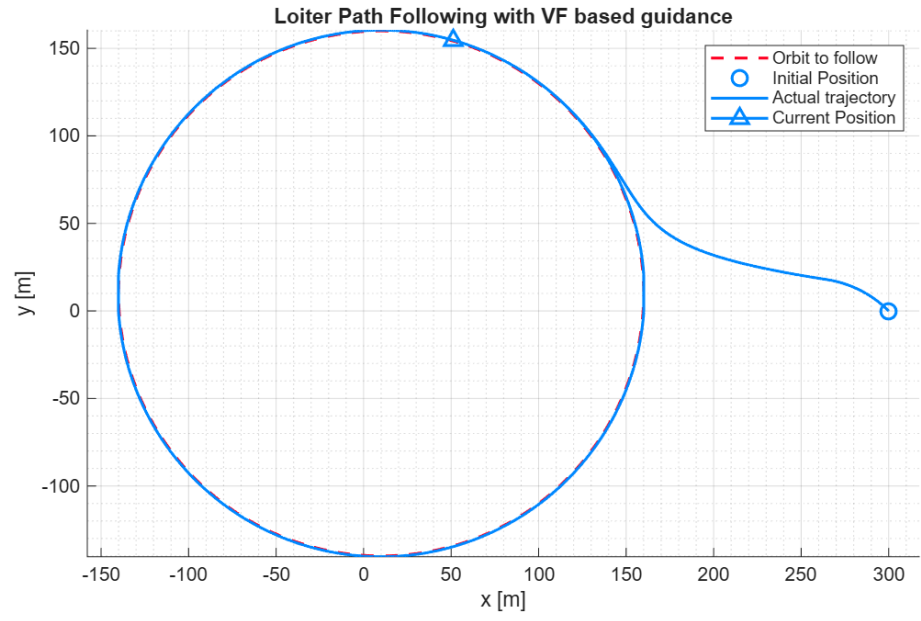
$$\dot{W} = \tilde{\chi} \left(\alpha(\chi^c - \chi) - \frac{V_g}{d} \sin(\chi - \gamma) + \beta V_g \cos(\chi - \gamma) \right)$$

again taking $\dot{\tilde{\chi}} = -\kappa \text{sign}(\tilde{\chi})$, this implies for $\kappa > 0$, $\dot{W} \leq -\kappa|\tilde{\chi}|$, from which we can conclude that $\tilde{\chi} \rightarrow 0$ in finite time.

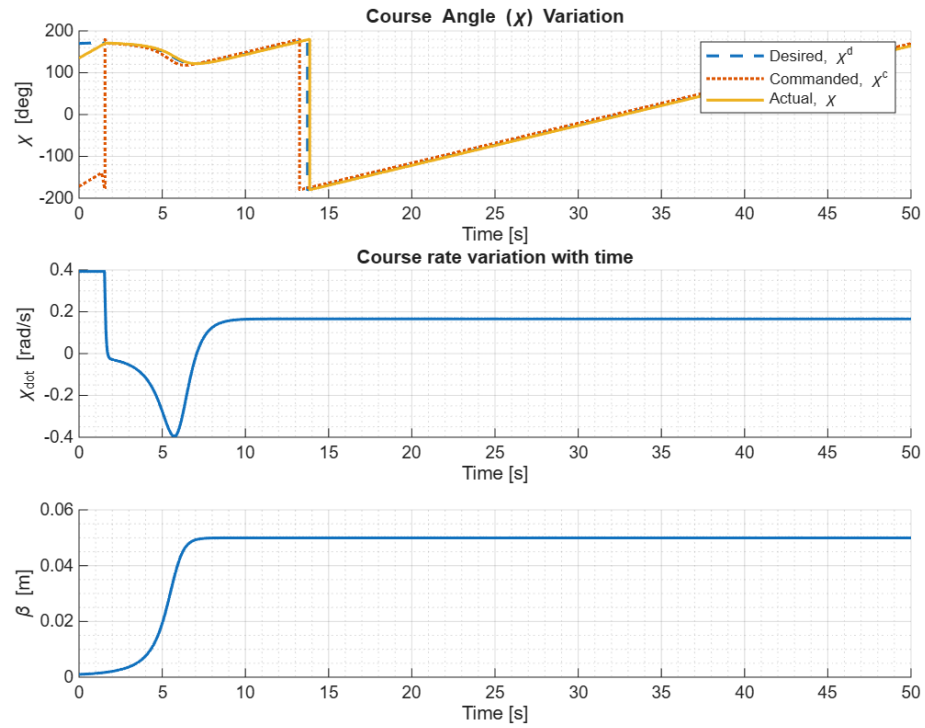
Again, to avoid chattering, the saturation function with the boundary region width of $\epsilon(> 0)$ is used in place of the sign function. Therefore, the control law,

$$\chi^c = \chi + \frac{V_g}{\alpha d} \sin(\chi - \gamma) - \frac{\beta}{\alpha} V_g \cos(\chi - \gamma) - \frac{\kappa}{\alpha} \text{sat} \left(\frac{\tilde{\chi}}{\epsilon} \right) \quad (2.26)$$

Simulation Similarly, for the Simulation of the loiter/orbit path following using the VF-based guidance, the loiter to follow is given as the center of the loiter $O(10, 10) m$ and the radius $150 m$. The UAV parameters such as constant airspeed $25 m/s$, initial position $(300, 0) m$, initial course angle $3\pi/4 rad$, course angle at infinity $\chi^\infty = \pi/2$, course hold controller gain ($\alpha =$) 1.65, Vector field convergence parameter ($k =$) 0.05, SMC gain ($\kappa =$) $\pi/2$ and width of the boundary region ($\epsilon =$) 0.1 and the maximum allowable roll angle is 45° .



(a) Trajectory



(b) Different parameters Variations with time while Loiter path following with VF guidance

FIGURE 2.18: Loiter/Orbit Path Following with VF-based guidance

2.2.2.2 Combined Vector Field (CVF) based guidance

The Combined Vector Field (CVF) guidance approach also belongs to the class of vector field-based guidance. However, unlike standard vector fields or Lyapunov vector fields, the CVF approach explicitly constructs the guidance field in two components. Based on the Helmholtz decomposition theorem, any vector field can be expressed as the sum of a solenoidal (divergence-free) component and a conservative (curl-free) component. In the context of CVF guidance, the solenoidal component induces motion tangential to the path, while the conservative component drives the convergence towards or away from the path. The approach and methodologies used by Liang et. al. in [2] on Combined Vector Field-based guidance will be discussed in **next chapter**.

Chapter 3

Combined Vector Field (CVF) Based Guidance

This chapter addresses the problem of following arbitrary twice-differentiable curves in both two-dimensional (2D) and three-dimensional (3D) spaces. Using the Helmholtz decomposition, guidance vector fields are synthesized by blending conservative and solenoidal components through a distance-based weighting relative to the reference path. A saturated course rate controller is then derived for the planar (2D) paths, and jointly saturated course and climb rate controllers are derived for spatial (3D) paths.

The following aspects also need to be emphasized.

- The 2D and 3D coordinates are denoted as $(x, y)^T$ and $(x, y, z)^T$ respectively.
- The 2D curve is defined as $C2 : f(x, y) = 0$ and 3D curve by a couple of equations $C3 : f(x, y, z) = 0, \quad g(x, y, z) = 0$ respectively.
- A 2D or 3D curve is said to be twice differentiable only if the first and second partial derivatives of the functions defining the curve exist.

3.1 2D Path Following

3.1.1 Problem Description

In the 2D path following, the UAV is considered to have altitude hold with zero climb rate and a velocity hold for constant airspeed (V_a). The kinematic model of the UAV in the inertial XY frame, as in figure 2.1, can be given as,

$$\dot{x} = V_a \cos(\psi) + w_x = V_g \cos(\chi) \quad (3.1)$$

$$\dot{y} = V_a \sin(\psi) + w_y = V_g \sin(\chi) \quad (3.2)$$

$$\dot{\chi} = u_1 \quad (3.3)$$

where, $\mathbf{p} = (x, y)^T$ is the position of the UAV in the 2D inertial frame, $\mathbf{v} = (\dot{x}, \dot{y})^T$ is the inertial velocities, ψ is the UAV heading angle, χ is the course angle, V_a is the constant airspeed, V_g is the ground speed, and $w = (w_x, w_y)^T$ is the wind velocities, u_1 is the commanded course rate satisfying the constraint $|u_1| \leq \dot{\chi}_{max}$.

Additionally, it can be observed that the relationship between airspeed and ground speed can be expressed as follows,

$$V_g = (w_x \cos \chi + w_y \sin \chi) + \sqrt{V_a^2 - (w_x \sin \chi - w_y \cos \chi)^2} \quad (3.4)$$

Now, given a desired 2D arbitrary twice differentiable curved path $C_2 : f(x, y) = 0$, we define the directional distance between the current UAV position $\mathbf{p}$ to C_2 as $r = f(\mathbf{p}) = f(x, y)$. Then, a large $|r|$ implies that the UAV is very far away from the curve C_2 , and $r = 0$ implies that the UAV is on the curve itself.

The 2D arbitrary twice-differentiable curved path following problem can be summarized as follows: To design the input command u_1 for the UAV whose kinematics are given by (3.1)-(3.3). Such that, (i) The input constraint $|u_1| \leq \dot{\chi}_{max}$ holds. (ii)

and if possible, force the UAV to fly along the curve C_2 , and if not, force it to make the absolute distance $|r|$ as small as possible.

3.1.2 2D Combined Vector Field Formulation

The Helmholtz theorem states that any arbitrary vector field can be decomposed into a conservative and a solenoidal part. The conservative part is irrotational (curl-free) in nature and forces the UAV away from or towards the desired path. The Solenoidal part is purely rotational, which forces the UAV to align in the direction of the curves in the desired path. The Conservative and solenoidal parts of the vector field can be determined by obtaining the gradient of the curve equation (∇f) and the curl of the curve equation ($\nabla \times f$), respectively.

Then the Conservative part after normalization,

$$\mathbf{v}_c = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (3.5)$$

and the solenoidal part after normalization,

$$\mathbf{v}_s = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_y \\ -f_x \end{pmatrix} \quad (3.6)$$

It can also be seen from (3.5) and (3.6) that both conservative and solenoidal fields are normal to each other at every point in space.

Now, according to Helmholtz's decomposition theorem, a certain combination of two perpendicular vector fields gives a new vector field. Furthermore, if we construct the new vector field from the conservative and solenoidal components such that, (i) when $\mathbf{p}$ is located very far away, i.e., $|r|$ is huge, then the new vector field aligns with the conservative field and forces the UAV to move directly towards the curve

C_2 . (ii) When $\mathbf{p}$ is on the curve C_2 , i.e., $r = 0$, then the new vector field is totally aligned with the solenoidal vector field, which forces the UAV to follow and remain on the curve C_2 .

Hence, the new combined vector field can be given as,

$$\mathbf{v}_d = \begin{pmatrix} \dot{x}_d \\ \dot{y}_d \end{pmatrix} = -V_g \tanh(\kappa r) \mathbf{v}_c + s V_g \text{sech}(\kappa r) \mathbf{v}_s \quad (3.7)$$

$$= \frac{V_g}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} -f_x \tanh(\kappa f(x, y)) + s f_y \text{sech}(\kappa f(x, y)) \\ -f_y \tanh(\kappa f(x, y)) - s f_x \text{sech}(\kappa f(x, y)) \end{pmatrix} \quad (3.8)$$

Where $\kappa > 0$ is a Vector field design parameter. The larger κ , the faster the convergence. Parameter $s = \pm 1$ determines the direction of the solenoidal vector field. The convergence of this derived combined vector field is proved in [2].

3.1.3 Controller Design

The saturated course rate controller for the constrained UAV based on the above-derived combined vector field. The desired course determined by the combined vector field in (3.8) can be computed as

$$\begin{aligned} \chi_d &= \arctan2(\dot{y}_d, \dot{x}_d) \\ &= \arctan2(-f_y \tanh(\kappa f) - s f_x \text{sech}(\kappa f), -f_x \tanh(\kappa f) + s f_y \text{sech}(\kappa f)) \end{aligned} \quad (3.9)$$

where $\arctan2(\cdot, \cdot) \in (-\pi, \pi]$. Again taking the course error $\chi_e = \langle \chi - \chi_d \rangle$, where $\langle \cdot \rangle$ is the angle wrapper.

Now, the time derivative of the desired course angle,

$$\dot{\chi}_d = \frac{\partial \chi_d}{\partial x} \dot{x} + \frac{\partial \chi_d}{\partial y} \dot{y} = V_g (A_x \cos(\chi) + A_y \sin(\chi)) \quad (3.10)$$

where

$$A_x = \frac{\partial \chi_d}{\partial x} = -s\kappa f_x \text{sech}(\kappa f) - \frac{f_{xx}f_y - f_x f_{xy}}{f_x^2 + f_y^2} \quad (3.11)$$

$$A_y = \frac{\partial \chi_d}{\partial y} = -s\kappa f_y \text{sech}(\kappa f) + \frac{f_{yy}f_x - f_y f_{xy}}{f_x^2 + f_y^2} \quad (3.12)$$

We can now design the saturated course rate controller for the path following of curve C_2 ,

$$u_1 = \begin{cases} \omega_1, & \text{if } |\omega_1| \leq \dot{\chi}_{max} \\ \text{sign}(\omega_1)\dot{\chi}_{max}, & \text{if } |\omega_1| > \dot{\chi}_{max} \end{cases} \quad (3.13)$$

where $\omega_1 = -k_\chi \chi_e + \dot{\chi}_d$. This is essentially a saturated PD Controller with a proportional positive feedback gain of k_χ .

Remark 1 If we do not include the course error in the controller, then the path following convergence will slow down, and in the worst case, it will not converge to the path at all. (Discussed in [2])

Remark 2 This combined Vector field approach will always converge to the path as it is globally feasible. (Discussed in [2])

Remark 3 Other feasibility conditions and stability analysis have been done for the 2D path following case in [2].

3.1.4 Simulations

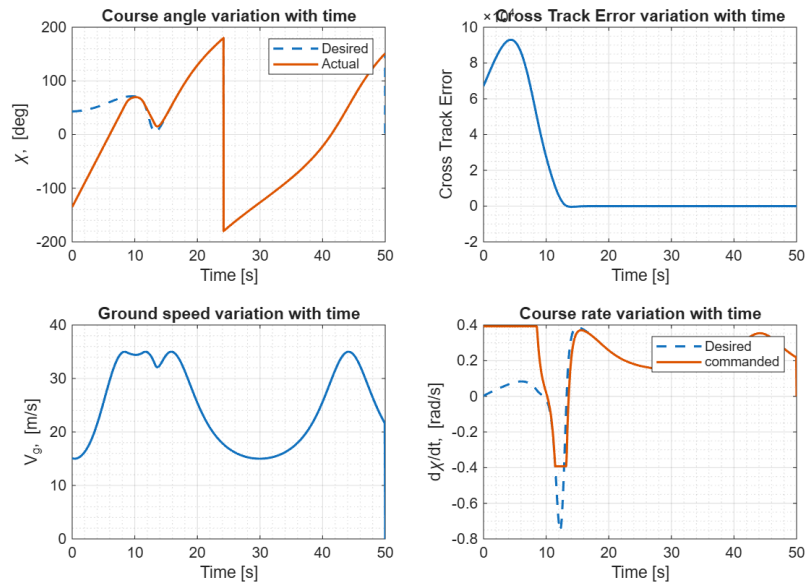
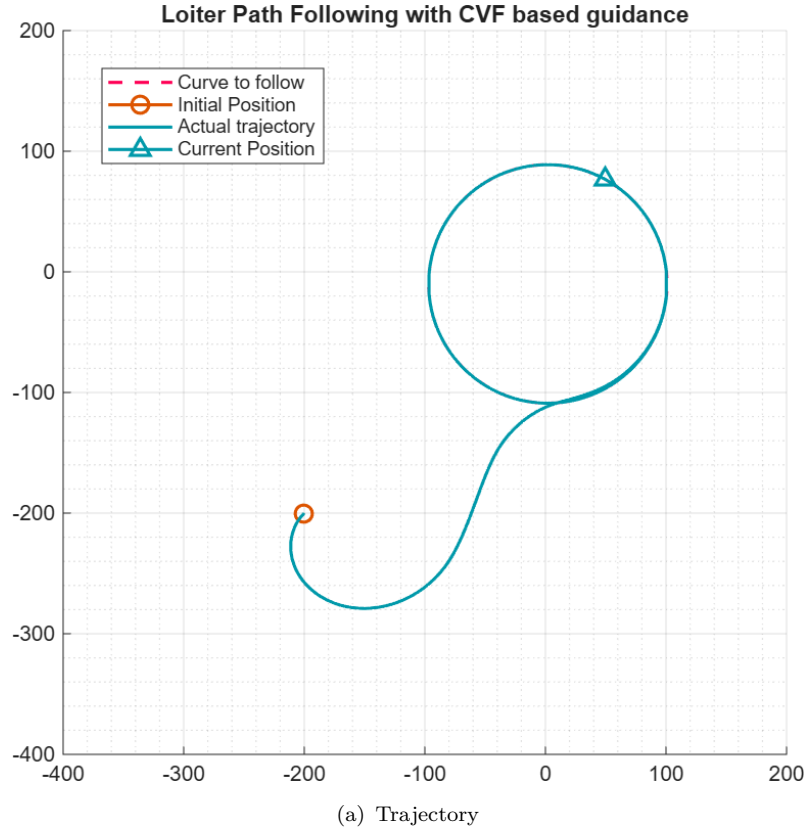
3.1.4.1 Loiter Path Following

For 2D Loiter Path following using CVF-based guidance, the loiter path is given by,

$$f(x, y) = (x - 2)^2 + (y + 10)^2 - 9775 \quad (3.14)$$

with constant airspeed 25 m/s , initial position $P_0(-200, -200) \text{ m}$, initial course angle $-3\pi/4$, Vector field design parameter ($\kappa_0 =$) 13, proportional feedback gain ($k_\chi =$

) 1.5 and the maximum allowable roll angle is 45° which limits the course rate. Additionally, simulation is done in the presence of wind $[6, 8] \text{ m/s}$.



(b) Different parameters Variations with time while Loiter path following with CVF guidance

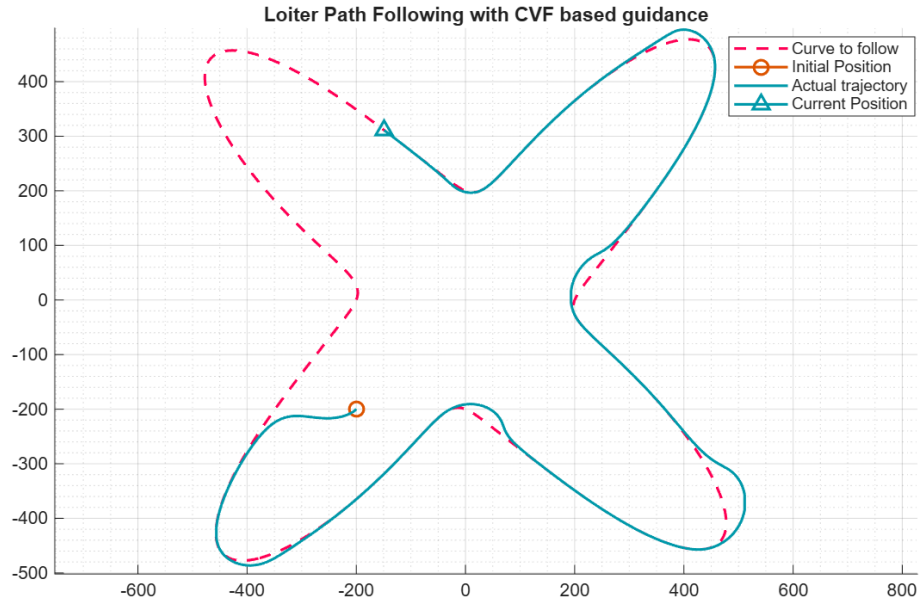
FIGURE 3.1: Loiter/Orbit Path Following with CVF-based guidance

3.1.4.2 Star Curve Path following

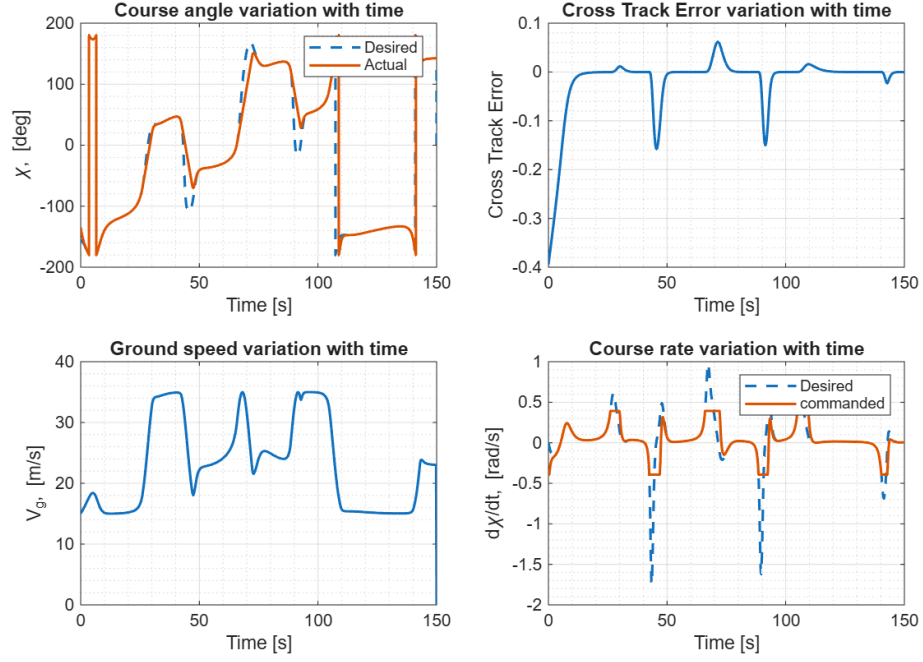
For a star curve path following with CVF guidance, only the curve to follow changes, and all other parameters remain the same. The Star curve can be given by,

$$f(x, y) = \frac{1}{(x^2 + y^2)^{\frac{9}{2}} + 1} (-a_0x^8 - 8b_0x^7y - 4a_0x^6y^2 + 56b_0x^5y^3 - 6a_0x^4y^4 - 56b_0x^3y^5 - 4a_0x^2y^6 + 8b_0xy^7 - a_0y^8 + (x^8 + x^6y^2 + x^4y^4 + x^2y^6 + y^8)\sqrt{x^2 + y^2}) \quad (3.15)$$

where $a_0 = 200$ and $b_0 = 10$.



(a) Trajectory



(b) Different parameters Variations with time while Star path following with CVF guidance

FIGURE 3.2: Star Curve Path Following with CVF-based guidance

3.2 3D Path Following

3.2.1 Problem Description

Similarly, as for the 2D case, we use the following simplified 3D kinematic model of the UAV in the inertial XYZ frame.

$$\dot{x} = V_a \cos(\theta) \cos(\psi) + w_x = V_g \cos(\gamma) \cos(\chi) \quad (3.16)$$

$$\dot{y} = V_a \cos(\theta) \sin(\psi) + w_y = V_g \cos(\gamma) \sin(\chi) \quad (3.17)$$

$$\dot{z} = V_a \sin(\theta) + w_z = V_g \sin(\gamma) \quad (3.18)$$

$$\dot{\chi} = u_1 \quad (3.19)$$

$$\dot{\gamma} = u_2 \quad (3.20)$$

where $\mathbf{p} = (x, y, z)^T$ is the position of the UAV in inertial space, $\mathbf{v} = (\dot{x}, \dot{y}, \dot{z})^T$ is the velocity of the UAV with respect to the inertial frame, V_a is the constant airspeed, θ

and ψ are the Pitch and Heading angle of the UAV respectively, $w = (w_x, w_y, w_z)^T$ is the wind velocity, $V_g = \|\mathbf{v}\|$ is the ground speed of the UAV (the speed as seen from the inertial frame/ground), $\chi \in (-\pi, \pi]$ and $\gamma \in [-\pi/2, \pi/2]$ are the Course and Climb angle of the UAV (measured w.r.t. air). u_1 is the course rate input, and u_2 is the climb rate input commanded from the controller. Input constraints are given as,

$$|u_1| \leq \dot{\chi}_{max}, \quad |u_2| \leq \dot{\gamma}_{max} \quad (3.21)$$

Remark The above formulation of the 3D kinematic model in terms of pitch (θ) and heading (ψ) is only valid if the velocity vector of the UAV is always aligned with the *+ve* body x-axis. This also means that the angle of attack and the side slip angle are always zero.

Also, the relationship between UAV airspeed and the ground speed according to the above formulation can be given as,

$$V_g = (w_x \cos \chi + w_y \sin \chi) \cos \gamma + \frac{1}{2} \sqrt{4V_a^2 - 3(w_x^2 + w_y^2) + (1 + 2 \cos(2\gamma))(w_x \cos \chi + w_y \sin \chi)^2 - (w_x \sin \chi - w_y \cos \chi)^2} \quad (3.22)$$

Again, given a desired 3D arbitrary twice differentiable path $C_3 : \{f(x, y, z) = 0 \ \& \ g(x, y, z) = 0\}$. We define a distance $r = \sqrt{f^2(\mathbf{p}) + g^2(\mathbf{p})}$, which gives an indirect measure of the closeness of the point $\mathbf{p}$ from the desired curve C_3 .

Then, the 3D arbitrary twice differentiable curved path following problem can be summarized as: To design the input commands u_1 and u_2 such that for the constrained UAV whose kinematics is given by Eq. (3.16)-(3.20) such that (i) The input constrained to the UAV holds, (ii) If possible make the UAV fly along the curve C_3 , and if not make the UAV to follow the curve C_3 as closely as possible (i.e., minimum cross track error r).

3.2.2 3D Combined Vector Field Formulation

Similarly, for the 2D path-following case, Helmholtz's decomposition theorem also holds in 3D. Therefore, the conservative and solenoidal parts of the vector field can be obtained by taking the gradient of the 3D curve C_3 and computing the tangential direction to the 3D curve C_3 , respectively. The conservative part of the vector field can be computed as,

$$\begin{aligned}\mathbf{v}_c &= \frac{\nabla(fg)}{|\nabla(fg)|} \\ &= \frac{1}{N_c} \begin{pmatrix} ff_x + gg_x \\ ff_y + gg_y \\ ff_z + gg_z \end{pmatrix}\end{aligned}\quad (3.23)$$

and the solenoidal part of the vector field can be computed as,

$$\begin{aligned}\mathbf{v}_s &= \frac{\nabla f \times \nabla g}{|\nabla f \times \nabla g|} \\ &= \frac{1}{N_s} \begin{pmatrix} f_y g_z - f_z g_y \\ f_z g_x - f_x g_z \\ f_x g_y - f_y g_x \end{pmatrix}\end{aligned}\quad (3.24)$$

where N_c and N_s are the normalization factors given by,

$$N_c = \sqrt{(ff_x + gg_x)^2 + (ff_y + gg_y)^2 + (ff_z + gg_z)^2} \quad (3.25)$$

$$N_s = \sqrt{(f_y g_z - f_z g_y)^2 + (f_z g_x - f_x g_z)^2 + (f_x g_y - f_y g_x)^2} \quad (3.26)$$

According to the combination concept given in 2D, a similar 3D vector field can be generated as,

$$\mathbf{v}_d = \begin{pmatrix} \dot{x}_d \\ \dot{y}_d \\ \dot{z}_d \end{pmatrix} = -V_g \tanh(\kappa r) \mathbf{v}_c + sV_g \text{sech}(\kappa r) \mathbf{v}_s \quad (3.27)$$

similarly, as in the 2D combined vector field in (3.8), $s = \pm 1$ determines the direction of the path following, and $\kappa > 0$ is a CVF design parameter. From this CVF, we can also see that if the UAV is very far away from the curve C_3 , then the vector field aligns with the conservative vector field, and if the UAV is on the curve C_3 , then the vector field aligns with the solenoidal vector field.

Again, the proof that if the Combined vector field determines the trajectories in (3.27) will asymptotically converge to the desired 3D curve, is given in [2].

3.2.3 Controller Design

The saturated course rate and climb rate controller for the constrained UAV based on the above-derived combined vector field. The desired course and climb determined by the combined vector field in (3.27) can be computed as

$$\chi_d = \arctan2(\dot{y}_d, \dot{x}_d) \quad (3.28)$$

$$\gamma_d = \arctan2\left(\dot{z}_d, \sqrt{\dot{x}_d^2 + \dot{y}_d^2}\right) \quad (3.29)$$

where all the symbols have the same meaning as previously described. Denote the course error as $\chi_e = \chi - \chi_d \in (-\pi, \pi]$ and climb error as $\gamma_e = \gamma - \gamma_d \in (-\pi, \pi]$. Now, the time derivative of the desired course angle and desired climb angle,

$$\begin{aligned} \dot{\chi}_d &= \frac{\partial \chi_d}{\partial x} \dot{x} + \frac{\partial \chi_d}{\partial y} \dot{y} + \frac{\partial \chi_d}{\partial z} \dot{z} \\ &= V_g (B_x \cos(\gamma) \cos(\chi) + B_y \cos(\gamma) \sin(\chi)) + B_z \sin(\gamma) \end{aligned} \quad (3.30)$$

$$\begin{aligned} \dot{\gamma}_d &= \frac{\partial \gamma_d}{\partial x} \dot{x} + \frac{\partial \gamma_d}{\partial y} \dot{y} + \frac{\partial \gamma_d}{\partial z} \dot{z} \\ &= V_g (C_x \cos(\gamma) \cos(\chi) + C_y \cos(\gamma) \sin(\chi)) + C_z \sin(\gamma) \end{aligned} \quad (3.31)$$

where B_x , B_y , B_z , C_x , C_y and C_z can be computed by the chain rule.

We can now design the saturated course rate and climb rate controller for the path

following of the desired curve C_3 ,

$$u_1 = \begin{cases} \omega_1, & \text{if } |\omega_1| \leq \dot{\chi}_{max} \\ \text{sign}(\omega_1)\dot{\chi}_{max}, & \text{if } |\omega_1| > \dot{\chi}_{max} \end{cases} \quad (3.32)$$

$$u_2 = \begin{cases} \omega_2, & \text{if } |\omega_2| \leq \dot{\gamma}_{max} \\ \text{sign}(\omega_2)\dot{\gamma}_{max}, & \text{if } |\omega_2| > \dot{\gamma}_{max} \end{cases} \quad (3.33)$$

where $\omega_1 = -k_\chi \chi_e + \dot{\chi}_d$ and $\omega_2 = -k_\gamma \gamma_e + \dot{\gamma}_d$. These are essentially saturated PD controllers with proportional positive feedback gains of k_χ and k_γ for the course and climb controllers, respectively.

Remark 1 Similar to the 2D case, if we do not include the course error and climb errors in the controller, then the path following convergence will slow down, and in the worst case, it will not converge to the path at all. (Discussed in [2])

Remark 2 This combined Vector field approach will always converge to the path as it is globally feasible if the initial flight path angles are aligned. (Discussed in [2])

Remark 3 Other feasibility conditions and stability analysis have been done for the 3D path following case in [2].

3.2.4 Simulations

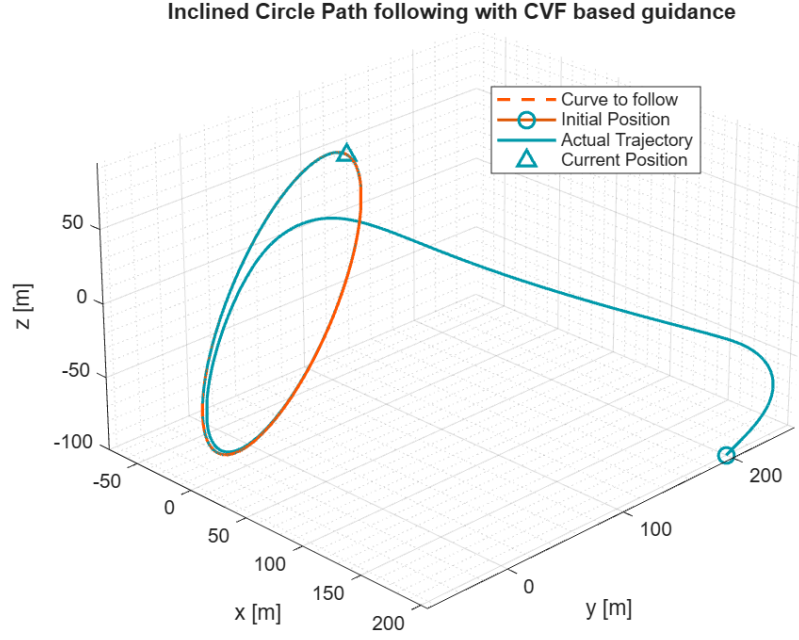
3.2.4.1 Inclined Circle Following

For 3D inclined circle Path following using CVF-based guidance, the equations of the surfaces that intersect, making the desired path, are given by,

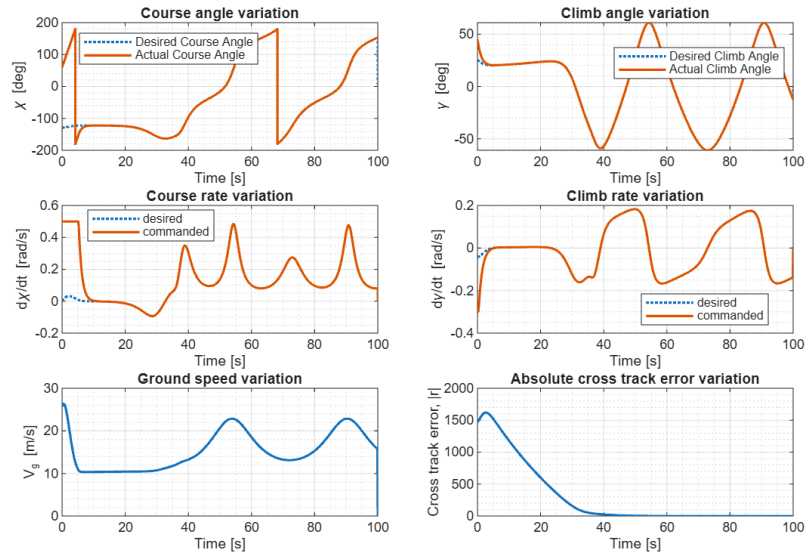
$$f(x, y, z) = \frac{1}{100} ((x - 2)^2 + (y + 10)^2 + (z - 15)^2 - 10000) \quad (3.34)$$

$$g(x, y, z) = 2x + 3y - 2z - 10 \quad (3.35)$$

with constant airspeed 20 m/s , initial position $P_0(200, 200, -100)^T \text{ m}$, initial course angle $\pi/3$, initial climb angle $\pi/4$, Vector field design parameter ($\kappa_0 =$) 10, proportional course and climb feedback gains ($k_\chi =$) 1 and ($k_\gamma =$) 1 respectively, maximum course rate 0.5 rad/s and maximum climb rate 0.3 rad/s . Additionally, simulation is done in the presence of wind $[6, 8, 0]^T \text{ m/s}$.



(a) Path following Trajectory



(b) Different parameters Variations with time

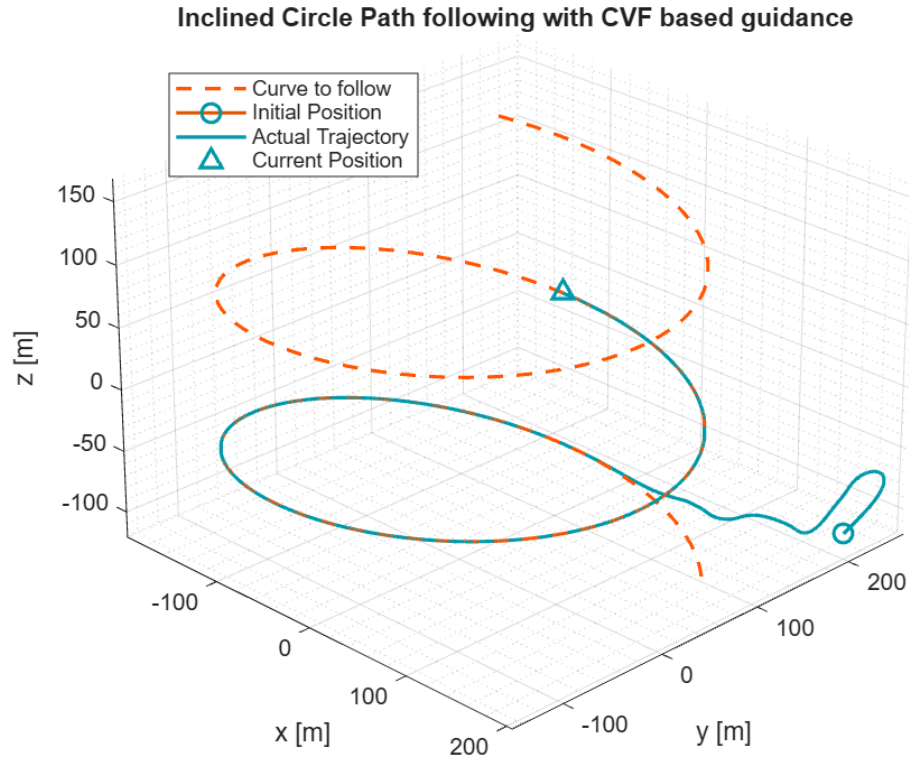
FIGURE 3.3: 3D Inclined Circle Path Following with CVF-based guidance

3.2.4.2 Helical Curve Path following

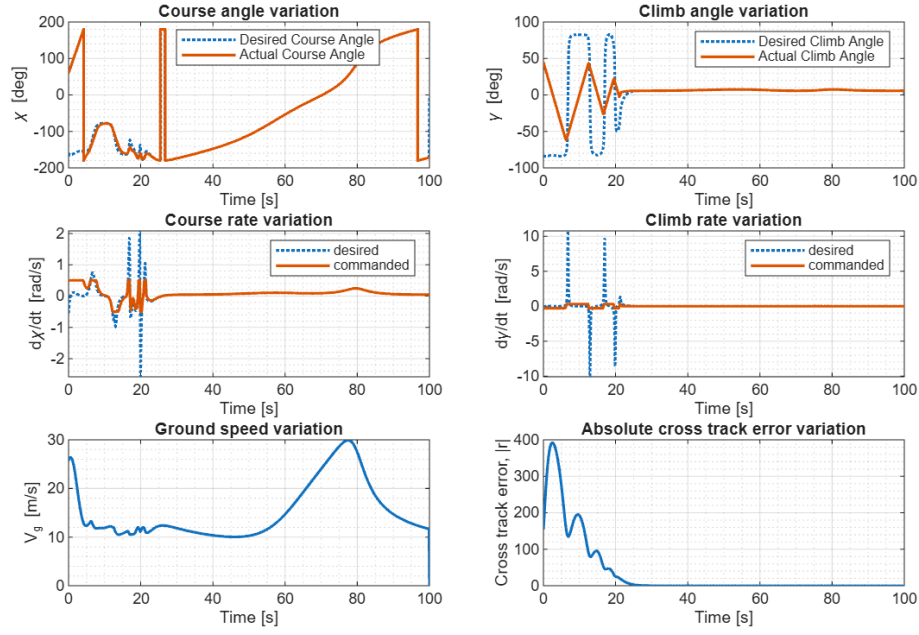
For a Helical curve path following with CVF guidance, only the curve to be followed and the corresponding design parameter κ_0 for the combined vector field change, and all other parameters remain the same. The design parameter is $\kappa_0 = 8$, and the helical curve can be given as,

$$f(x, y, z) = x - 200 \cos\left(\frac{z}{20}\right) \quad (3.36)$$

$$g(x, y, z) = y - 150 \sin\left(\frac{z}{20}\right) \quad (3.37)$$



(a) Path Following Trajectory



(b) Different parameters Variations with time

FIGURE 3.4: 3D Helical Curve Path Following with CVF-based guidance

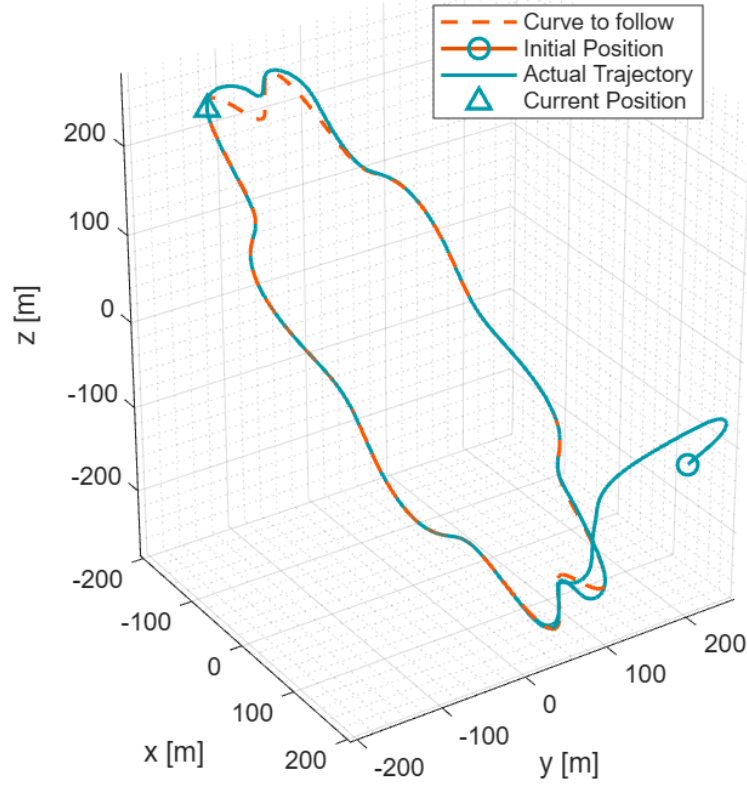
3.2.4.3 Inclined Star Curve Path following

For an inclined star curve path following with CVF guidance, only the curve to be followed and the corresponding design parameter κ_0 for the combined vector field change, and all other parameters remain the same. The design parameter is $\kappa_0 = 5$, and the helical curve can be given as,

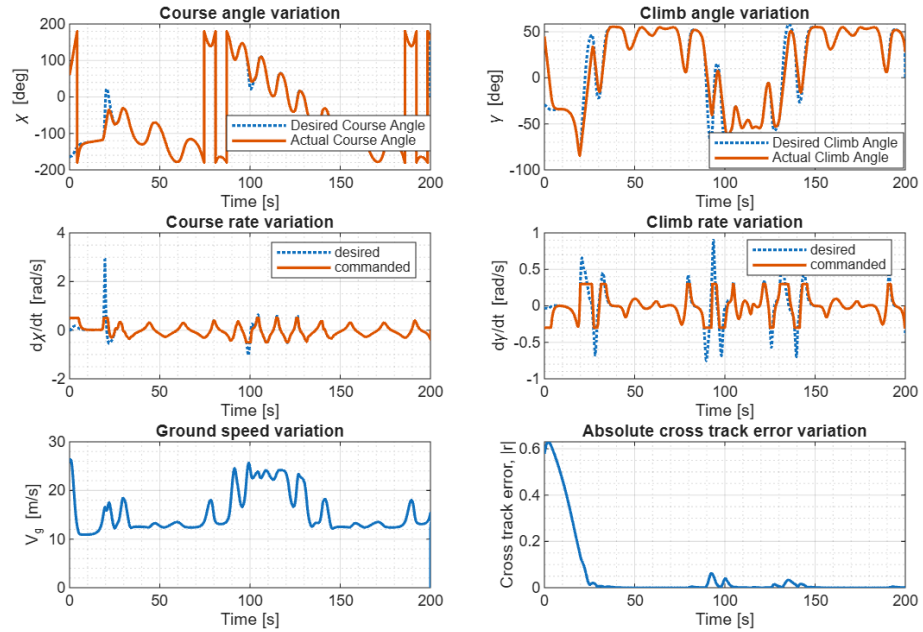
$$f(x, y, z) = \frac{x + y + z}{\sqrt{5x^2 + xy + 5y^2} + 1} \quad (3.38)$$

$$g(x, y, z) = \frac{1}{(5x^2 + xy + 5y^2)^{9/2} + 1} \left(-(5x^2 + xy + 5y^2)^{\frac{9}{2}} + 300\sqrt{2}(5x^2 + xy + 5y^2)^4 \right. \\ \left. + 45\sqrt{22}(x^2 - y^2)(x^2 + 20xy + y^2)(49x^4 - 20x^3y - 300x^2y^2 - 20xy^3 + 49y^4) \right) \quad (3.39)$$

Inclined Circle Path following with CVF based guidance



(a) Path Following Trajectory



(b) Different parameters Variations with time

FIGURE 3.5: 3D Inclined Star Curve Path Following with CVF-based guidance

3.3 Limitations

The Combined Vector Field (CVF) approach, being very successful in following the 2D or 3D curved path, has several limitations:

- No common design parameter κ can be found such that it generates the appropriate vector field for path following for any twice differentiable curve. This is because the cross-track error r is not a scale-free quantity, i.e., a curve can be defined as $f(x, y) = 0$ and $\lambda f(x, y) = 0$, where λ is a scalar scaling factor. The cross-track error for the same curve defined by different equations is $r = f(x, y)$ and $r = \lambda f(x, y)$, respectively. Hence, for the same curve with different curve equations, the resulting cross-track error r is different, and these cross-track errors cause the system to behave differently.

To mitigate this, as suggested in [2], we can introduce a parameter κ_0 which is defined as $\kappa = \kappa_0/|r_0|$, where r_0 is the cross-track error at the initial position.

In this way, we can normalize the whole κr term.

- In 3D case, the cross-track error is defined as $r = \sqrt{f^2 + g^2}$. Now, again, if we take two surfaces $f(x, y, z)$ and $g(x, y, z)$, such that in the domain around which the UAV will operate, value of the function $f(x, y, z)$ and $g(x, y, z)$ differs greatly. This implies that the UAV will be attracted to the curve with the higher contribution to the cross-track and will get stuck in that surface instead of converging to the desired curve. To mitigate this, again suggested in [2], we can compute the functional values at the initial position, and get a factor λ such that the cross-track error becomes $r = \sqrt{f^2 + (\lambda g)^2}$, where the contributing terms f and λg contributes to the cross-track error in equal proportions.
- Additionally, as noted in [2], the proposed guidance with combined vector fields cannot be easily extended for trajectory tracking, as it relies on reference tracking.

Chapter 4

Validation of Combined Vector Field Based Path Following with Quadrotor Dynamics

In the previous chapter, we formalized the Combined Vector Field (CVF)- based path following for arbitrary twice-differentiable curves as described in [2]. The simulations in that chapter have only been done with the kinematic model of the UAV. In this chapter, we formalize and simulate the full quadrotor dynamics using Sliding Mode Controllers (SMC).

4.1 Quadrotor Dynamics

The Quadrotor, as the name suggests, has four propellers at four corners of the vehicle, each rotating in the opposite direction. These four propellers can be controlled separately. For being in hover state, all four propellers spin at the same rotational speed to give a net upward force directly away from the force of gravity with no net moments. Now, to have six degrees of motion with the Quadrotor, different

combinations of the rotational speeds for four different rotors are considered.

To formalize the quadrotor dynamics, we first need to establish the coordinate system. Let $EXYZ$ axes, as in the figure 4.1, be the inertial set of axes and $Oxyz$ be the axes attached to the Quadrotor itself. Also note that the force of gravity acts on the Quadrotor in the $-ve$ z -axis. The subscript or superscript b will mean body axes, and i will mean the inertial axes later on in the discussion.

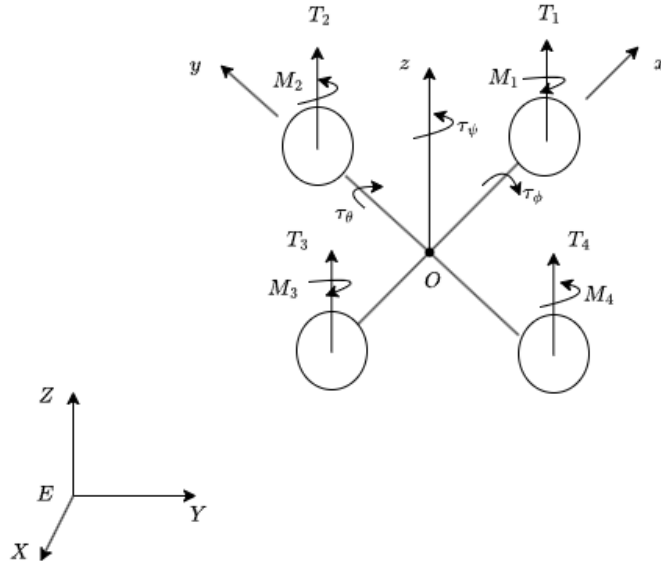


FIGURE 4.1: Schematic of Quadrotor Forces and Moments with the Frame of References

Now, from Newton's second law, writing the forces on the Quadrotor,

$$\begin{aligned}\vec{\mathbf{F}}^b &= m \frac{d}{dt} \Big|_i \vec{\mathbf{v}}^b \\ &= m \left(\frac{d}{dt} \Big|_b \vec{\mathbf{v}}^b + \vec{\omega}_{bi}^b \times \vec{\mathbf{v}}^b \right)\end{aligned}$$

upon rearranging,

$$\frac{d}{dt} \Big|_b \vec{\mathbf{v}}^b = \frac{1}{m} \vec{\mathbf{F}}^b - \vec{\omega}_{bi}^b \times \vec{\mathbf{v}}^b$$

$$\begin{bmatrix} \dot{u} \\ \dot{v} \\ \dot{w} \end{bmatrix} = R_i^b \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix} - \vec{\omega}_{bi}^b \times \vec{v}^b + \begin{bmatrix} 0 \\ 0 \\ \frac{U_1}{m} \end{bmatrix} \quad (4.1)$$

Also, from the rigid body dynamics, the moments on the Quadrotor are the inertial time rate of change in the angular momentum of the Quadrotor resolved in the body frame. This implies,

$$\begin{aligned} \begin{bmatrix} \tau_\phi \\ \tau_\theta \\ \tau_\psi \end{bmatrix} &= \frac{d}{dt} \Big|_i \vec{H}^b = \frac{d}{dt} \Big|_i (I \vec{\omega}_{bi}^b) \\ &= \frac{d}{dt} \Big|_b (I \vec{\omega}_{bi}^b) + \vec{\omega}_{bi}^b \times (I \vec{\omega}_{bi}^b) \end{aligned}$$

upon rearranging,

$$\begin{aligned} \frac{d}{dt} \Big|_b \vec{\omega}_{bi}^b &= I^{-1} [\vec{\tau}^b - \vec{\omega}_{bi}^b \times (I \vec{\omega}_{bi}^b)] \\ \begin{bmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{bmatrix} &= I^{-1} [\vec{\tau}^b - \vec{\omega}_{bi}^b \times (I \vec{\omega}_{bi}^b)] \end{aligned} \quad (4.2)$$

and,

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = C_\theta^\omega \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix} \quad (4.3)$$

where,

$$R_b^i = \begin{bmatrix} \cos(\theta) \cos(\psi) & \sin(\phi) \sin(\theta) \cos(\psi) & \cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi) \\ \cos(\theta) \sin(\psi) & \sin(\phi) \sin(\theta) \sin(\psi) & \cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi) \\ -\sin(\theta) & \sin(\phi) \cos(\theta) & \cos(\phi) \cos(\theta) \end{bmatrix} \quad (4.4)$$

$$C_\theta^\omega = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\phi) & \sin(\phi) \cos(\theta) \\ 0 & -\sin(\phi) & \cos(\phi) \cos(\theta) \end{bmatrix} \quad (4.5)$$

m is the mass of the Quadrotor.

$\vec{v}^b$ is the velocity of the Quadrotor resolved in the body frame ($Oxyz$ frame) ($= [u, v, w]^T$)

$\frac{d}{dt}|_i$ is the time rate of change in the inertial frame.

$\frac{d}{dt}|_b$ is the time rate of change in the body frame.

$\vec{F}^b$ is the force on the Quadrotor resolved in the body frame.

$\vec{\omega}_{bi}^b$ is the angular velocity of the body frame with respect to the inertial frame resolved in the body frame. ($= [p, q, r]^T$)

g is the gravitational acceleration. ($= 9.81 \text{ m/s}^2$)

R_i^b is the Transformation matrix, which transforms the quantities from the inertial frame to the body frame.

U_1 is the total upward thrust by all the propellers in the body z -direction to the Quadrotor. (see Eqns. (4.7))

τ_ϕ is the moment on the Quadrotor which makes the quadrotor roll along the body x -axis.

τ_θ is the moment on the Quadrotor, which makes the quadrotor pitch along the body y -axis.

τ_ψ is the moment on the Quadrotor, which makes the quadrotor yaw along the body z -axis.

$\vec{H}^b$ is the angular momentum of the Quadrotor resolved in body axes.

I is the Inertia matrix, as the Quadrotor considered in our case is symmetric in xy , yz , and xz planes passing through the centroid of the Quadrotor. Hence, the

Inertia matrix can be given as,

$$I = \begin{bmatrix} I_x & 0 & 0 \\ 0 & I_y & 0 \\ 0 & 0 & I_z \end{bmatrix} \quad (4.6)$$

C_θ^ω is the Transformation matrix which transforms the attitude rates (roll, pitch, yaw rates) to angular velocities.

The motor equations can also be written for the Quadrotor in figure 4.1. The thrust and drag coefficients of the propeller are b and d , respectively. And the thrust forces (T_i) and the moment due to drag (M_i) can be given in terms of the propeller rotation speed (Ω) as $T_i = b\Omega_i^2$ and $M_i = d\Omega_i^2$. The motor equations can then be given as,

$$T = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2)$$

$$\tau_\phi = bl(-\Omega_2^2 + \Omega_4^2)$$

$$\tau_\theta = bl(-\Omega_1^2 + \Omega_3^2)$$

$$\tau_\psi = d(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2)$$

again, rearranging in terms of the input variable U_1 , U_2 , U_3 and U_4 ,

$$U_1 = T = b(\Omega_1^2 + \Omega_2^2 + \Omega_3^2 + \Omega_4^2) \quad (4.7)$$

$$U_2 = \frac{\tau_\phi}{l} = b(-\Omega_2^2 + \Omega_4^2) \quad (4.8)$$

$$U_3 = \frac{\tau_\theta}{l} = b(-\Omega_1^2 + \Omega_3^2) \quad (4.9)$$

$$U_4 = \tau_\psi = d(-\Omega_1^2 + \Omega_2^2 - \Omega_3^2 + \Omega_4^2) \quad (4.10)$$

The above-derived dynamical model of the Quadrotor is highly accurate, with very few assumptions made. This model is hence used to determine the next state of the Quadrotor in the simulation, given its current state and inputs. However, to design the controllers, the simplified version of the above dynamical model is used, which

can be given in terms of the equations given below,

$$\ddot{x} = \frac{U_1}{m}(\cos(\phi) \sin(\theta) \cos(\psi) + \sin(\phi) \sin(\psi)) \quad (4.11)$$

$$\ddot{y} = \frac{U_1}{m}(\cos(\phi) \sin(\theta) \sin(\psi) - \sin(\phi) \cos(\psi)) \quad (4.12)$$

$$\ddot{z} = -g + \frac{U_1}{m} \cos(\phi) \cos(\theta) \quad (4.13)$$

$$\ddot{\phi} = \dot{\phi} \dot{\psi} \left(\frac{I_y - I_z}{I_x} \right) + \frac{l}{I_x} U_2 \quad (4.14)$$

$$\ddot{\theta} = \dot{\phi} \dot{\psi} \left(\frac{I_z - I_x}{I_y} \right) + \frac{l}{I_y} U_3 \quad (4.15)$$

$$\ddot{\psi} = \dot{\phi} \dot{\theta} \left(\frac{I_x - I_y}{I_z} \right) + \frac{1}{I_z} U_4 \quad (4.16)$$

4.2 Quadrotor Control

The controller used to control the quadrotor is a Sliding Mode Controller. Sliding Mode Control is a technique that originates from Variable Structure Control (VSC). The Sliding Mode Control objective involves two parts. The first part is a control law which forces the error vector towards a decision rule, called *Sliding Surface*. In the second part, once the error vector is restricted to the sliding surface, it tracks the dynamics imposed by the equations describing the sliding surface. This second part of the controller is referred to as the *Equivalent Control*. The Sliding surface of the system is chosen as,

$$s = \dot{e} + \lambda e \quad (4.17)$$

where $\lambda > 0$ is a tuning parameter and e is the error vector.

The control law $U(t)$ is composed of two parts, as discussed earlier: a continuous part $u_{eq}(t)$ and a discontinuous part $u_D(t)$. [14]

$$U(t) = u_{eq}(t) + u_D(t) \quad (4.18)$$

The first part of this equation (equivalent control) can be determined by imposing the condition where the system is on the sliding surface s , then the sliding condition is $\dot{s} = 0$. And for the second part, which includes a non-linear switching element responsible for reaching the sliding surface, can be given as [14]

$$u_D(t) = k_D \frac{s(t)}{|s(t)| + \delta} \quad (4.19)$$

where k_D is a tunable parameter that is responsible for reaching the sliding surface. And δ is also a tunable parameter which helps reduce the chattering effects of a Sliding Mode Controller.

Now, for designing the altitude hold controller with SMC, define the altitude error,

$$e_1 = z_d - z \quad (4.20)$$

where z_d is the desired state and z is the current state. The sliding surface then can be obtained from Eq. (4.17),

$$s_1 = (\dot{z}_d - \dot{z}) + \lambda(z_d - z) \quad (4.21)$$

By applying the sliding condition $\dot{s} = 0$,

$$\dot{s} = (\ddot{z}_d - \ddot{z}) + \lambda(\dot{z}_d - \dot{z}) = 0$$

substituting (4.13),

$$\begin{aligned} \ddot{z}_d + g - \frac{U_{1eq}}{m} \cos(\phi) \cos(\theta) + \lambda_1(\dot{z}_d - \dot{z}) &= 0 \\ U_{1eq} &= \frac{m}{\cos(\phi) \cos(\theta)} [\ddot{z}_d + g + \lambda_1(\dot{z}_d - \dot{z})] \end{aligned} \quad (4.22)$$

Now, since the full control law is given as in (4.18) and the switching control is given in (4.19),

$$U_1 = \frac{m}{\cos(\phi) \cos(\theta)} [\ddot{z}_d + g + \lambda_1(\dot{z}_d - \dot{z})] + k_{D1} \frac{s_1}{|s_1| + \delta} \quad (4.23)$$

Similarly, for roll, pitch, and yaw control,

$$U_2 = \frac{I_x}{l} \left[\ddot{\phi}_d - \dot{\phi} \dot{\psi} \left(\frac{I_y - I_z}{I_x} \right) + \lambda_2(\dot{\phi}_d - \dot{\phi}) \right] + k_{D2} \frac{s_2}{|s_2| + \delta} \quad (4.24)$$

$$U_3 = \frac{I_y}{l} \left[\ddot{\theta}_d - \dot{\phi} \dot{\psi} \left(\frac{I_z - I_x}{I_y} \right) + \lambda_3(\dot{\theta}_d - \dot{\theta}) \right] + k_{D3} \frac{s_3}{|s_3| + \delta} \quad (4.25)$$

$$U_4 = I_z \left[\ddot{\psi}_d - \dot{\phi} \dot{\theta} \left(\frac{I_x - I_y}{I_z} \right) + \lambda_4(\dot{\psi}_d - \dot{\psi}) \right] + k_{D4} \frac{s_4}{|s_4| + \delta} \quad (4.26)$$

Since, in the case of a quadrotor, we can move to any location in 3D space by adjusting the roll (ϕ) angle and pitch (θ) angle without affecting the yaw (ψ) angle. This implies that the yaw angle is a free angle that can be set to any angle without affecting the whole system.

Now, in this quadcopter control problem, let the desired yaw angle (ψ_d) be equal to the course angle (χ). (or just the yaw angle is known)

Since, from the Eqs. (4.11) and (4.12), assume,

$$U_x = \cos(\phi) \sin(\theta) \cos(\psi_d) + \sin(\phi) \sin(\psi_d) \quad (4.27)$$

$$U_y = \cos(\phi) \sin(\theta) \sin(\psi_d) - \sin(\phi) \cos(\psi_d) \quad (4.28)$$

upon doing (4.27) $\times \sin(\psi_d)$ + (4.28) $\times \cos(\psi_d)$

$$\phi_d = \sin^{-1} (U_x \sin(\psi_d) - U_y \cos(\psi_d)) \quad (4.29)$$

similarly, upon doing (4.27) $\times \cos(\psi_d)$ + (4.28) $\times \sin(\psi_d)$

$$\theta_d = \sin^{-1} \left(\frac{U_x \cos(\psi_d) + U_y \sin(\psi_d)}{\cos(\phi_d)} \right) \quad (4.30)$$

Similar to the previous controller, the x and y controllers, then can be obtained as,

$$U_x = \frac{m}{U_1} [\ddot{x}_d + \lambda_x (\dot{x}_d - \dot{x})] + k_{Dx} \frac{s_x}{|s_x| + \delta} \quad (4.31)$$

$$U_y = \frac{m}{U_1} [\ddot{y}_d + \lambda_y (\dot{y}_d - \dot{y})] + k_{Dy} \frac{s_y}{|s_y| + \delta} \quad (4.32)$$

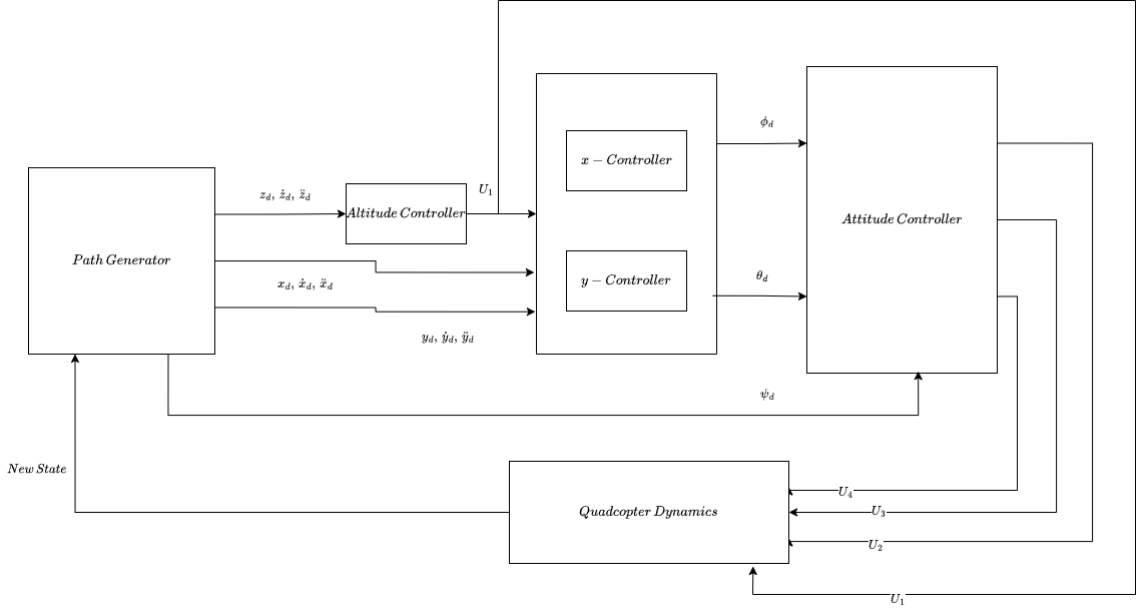


FIGURE 4.2: Schematic of the Control Architecture of the Quadrotor with Sliding Mode Controllers

4.3 Simulations

The Quadrotor dimensions and parameters are chosen to be as for **Q250 Quadrotor**. The physical dimensions of the Quadrotor are given as follows, mass of the Quadcopter is 444 g , arm length (from the centroid to one of the motor end) is 17.7 cm , the weight distribution and the overall structure of the Quadrotor is such that the moment of inertia along the axis passing through centroid and the two motors is $(I_x = I_y =)1.758 \times 10^{-3}\text{ Kg/m}^2$ and the moment of inertia along the direction perpendicular to the x and y axis is $(I_z =)3.492 \times 10^{-3}\text{ Kg/m}^2$, thrust coefficient is $9.92 \times 10^{-7}\text{ N}^2 \cdot \text{s}^2/\text{rad}^2$, drag coefficient is $1.49 \times 10^{-8}\text{ N}^2 \cdot \text{s}^2/\text{rad}^2$, maximum propeller angular velocity is 18510 RPM , maximum roll angle is 60° , and maximum climb angle is 60° .

The parameters involving the Quadrotor Sliding Mode Controller for full Quadrotor control are listed below in the table.

TABLE 4.1: Sliding Mode Controller gains for the Full Quadrotor Control

Parameter	z	x	y	ϕ	θ	ψ
k_D	5	0.35	0.35	2.5	2.5	1.2
λ	0.5	1.5	1.5	5	5	0.7
δ	0.3	0.3	0.3	0.3	0.3	0.3

Where k_D , λ , and δ have the same meaning as discussed in the Sliding Mode Controller Formulations.

The parameters involving the path generator (in this case Combined Vector Field (CVF) based guidance) are as follows, the constant airspeed of the Quadrotor is 10 m/s , wind velocity is $[3, 4, 0]\text{ m/s}$, the maximum course rate is 0.5 rad/s , the maximum climb rate is 0.3 rad/s , the proportional course feedback and climb feedback gains are 1 each.

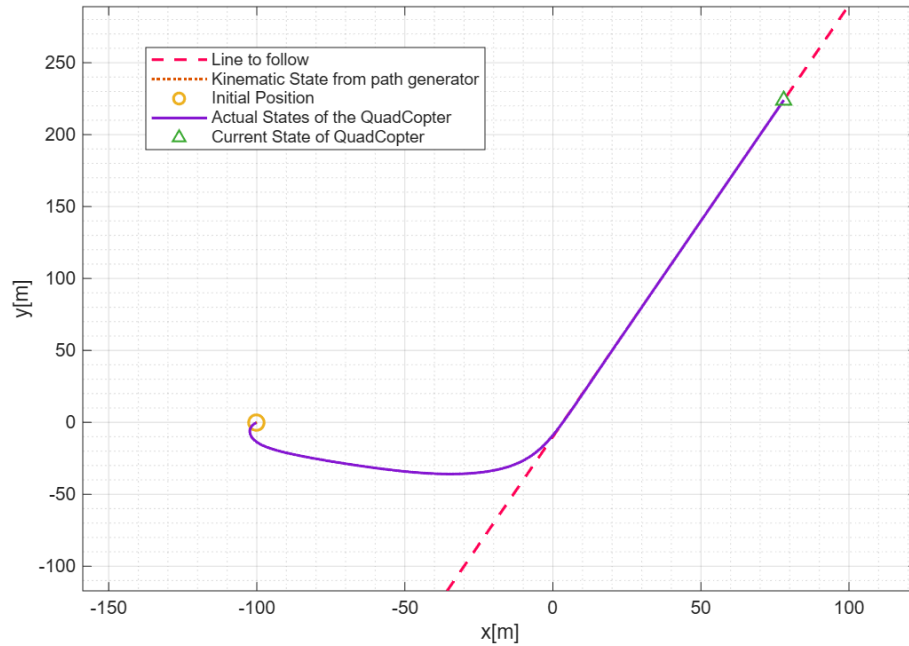
4.3.1 2D Simulations

4.3.1.1 Line Path Following

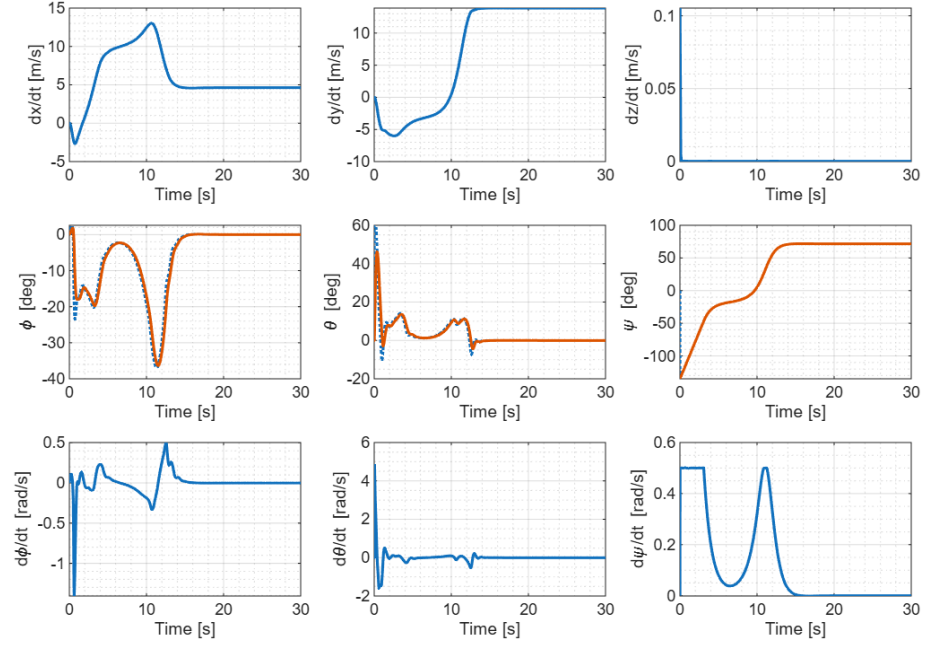
For line Path Following using CVF-based guidance with previously described Quadrotor dynamics, the line to follow is given by the curve,

$$f(x, y) = 500(y - 3x + 10) \quad (4.33)$$

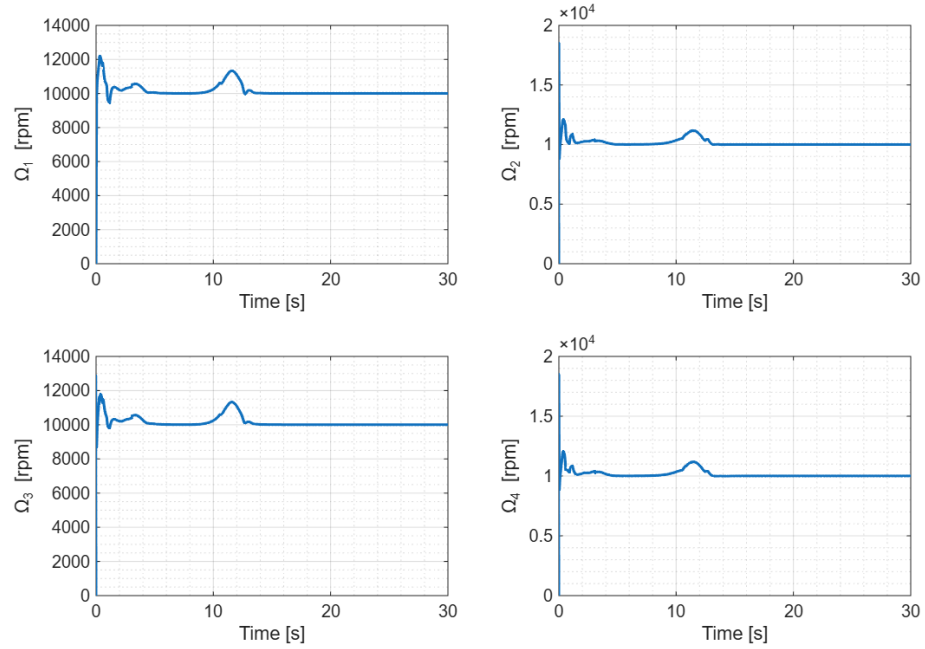
The initial state of the Quadrotor is given as, initial position is $P0(-100, 0) m$, initial course angle is $-3\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 8. **Remark** Notice the factor of the 500 multiplied to the curve itself. The multiplication of a factor does not change the curve itself; however, it changes the convergence of the method. It happens because $r = f(x, y)$ and the convergence is directly dependent on r . Similar phenomenon can also be seen in the 3D cases as well.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

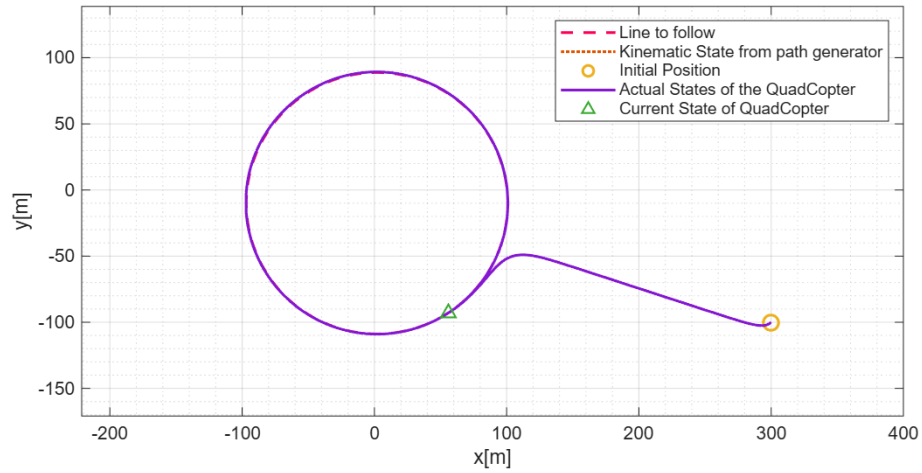
FIGURE 4.3: 2D Line Path Following with Quadrotor Dynamics with CVF-based guidance

4.3.1.2 Loiter Path Following

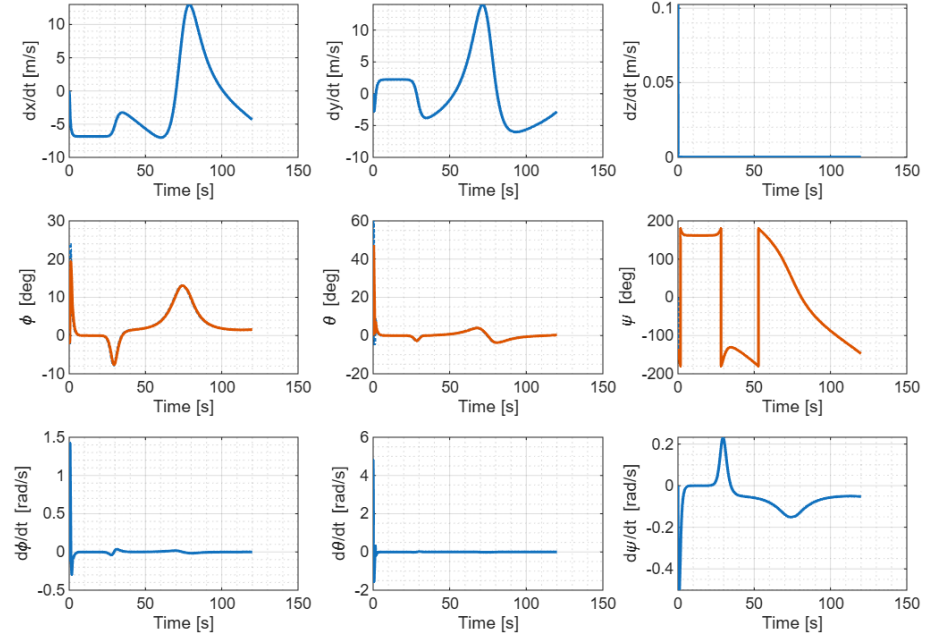
For 2D loiter Path Following using CVF-based guidance with Quadrotor dynamics, the Loiter to follow is given by the curve,

$$f(x, y) = 10((x - 2)^2 + (y + 10)^2 - 9775) \quad (4.34)$$

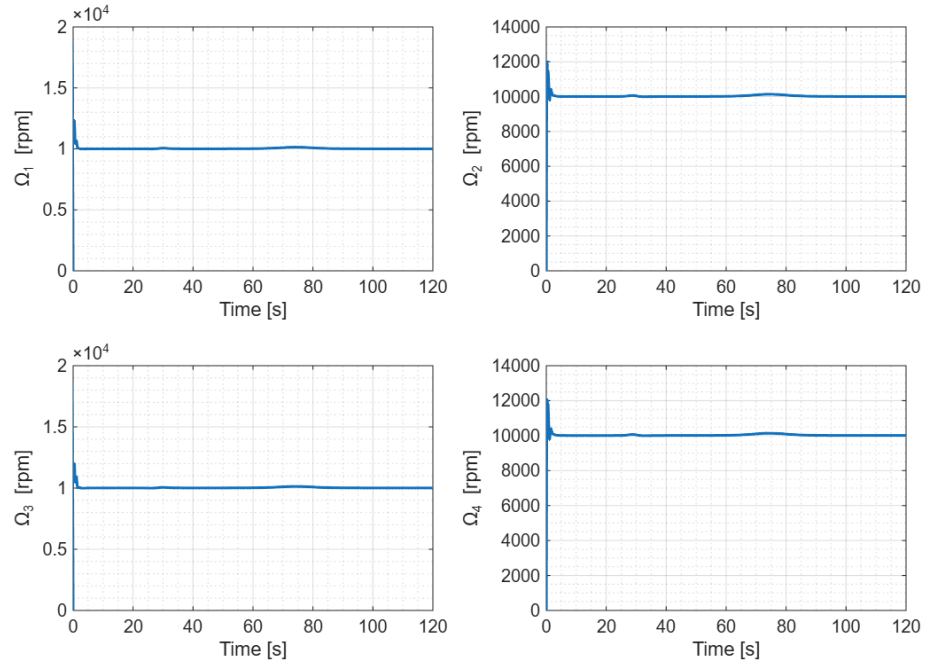
The initial state of the Quadrotor is given as: initial position is $P0(300, -100) m$, initial course angle is $-3\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 5.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

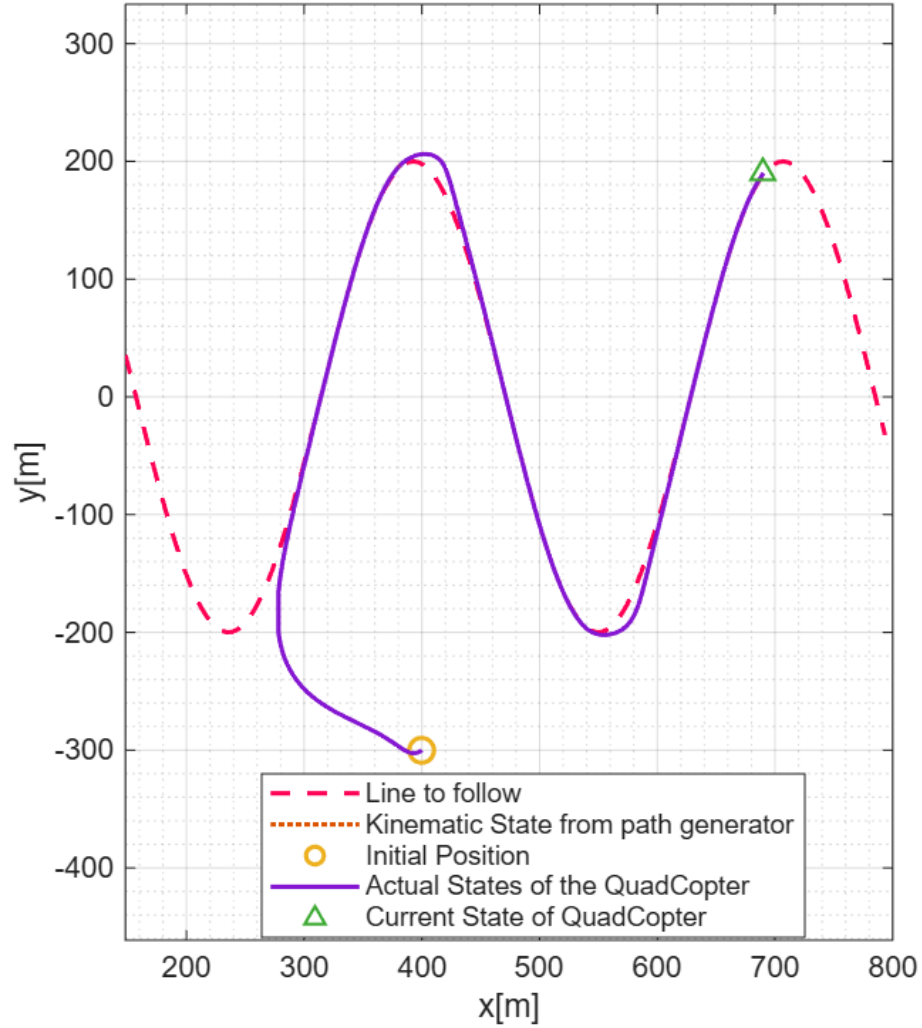
FIGURE 4.4: 2D Loiter Path Following with Quadrotor Dynamics with CVF-based guidance

4.3.1.3 Sine Curve Path Following

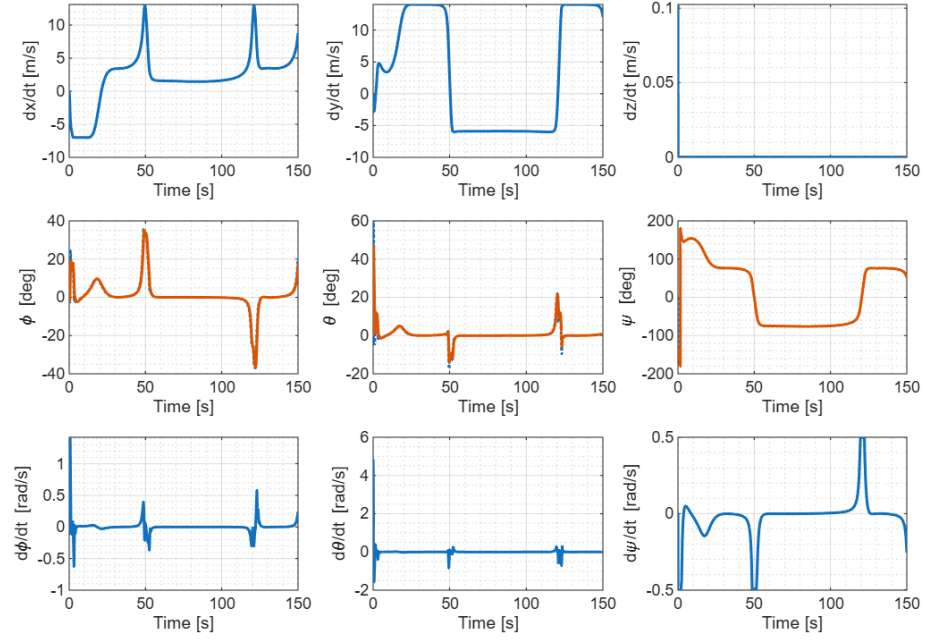
For 2D Sine Curve Path Following using CVF-based guidance with Quadrotor dynamics, the Sine Curve to follow is given by the curve,

$$f(x, y) = 150 \left(y - 200 \sin \left(\frac{x}{50} \right) \right) \quad (4.35)$$

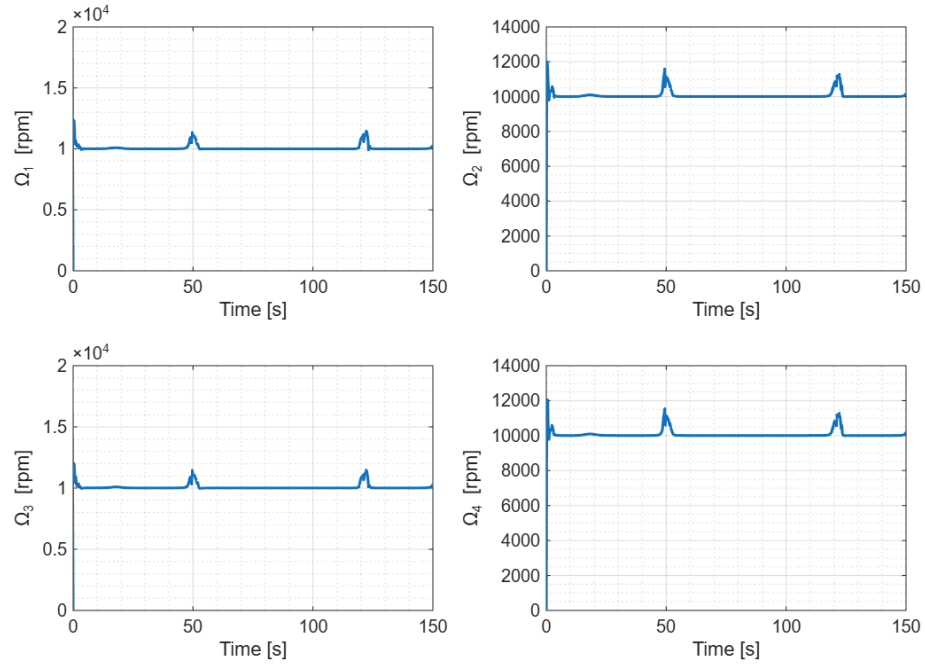
The initial state of the Quadrotor is given as: initial position is $P_0(400, -300) m$, initial course angle is $-3\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 5.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

FIGURE 4.5: 2D Sine Curve Path Following with Quadrotor Dynamics with CVF-based guidance

4.3.2 3D Simulations

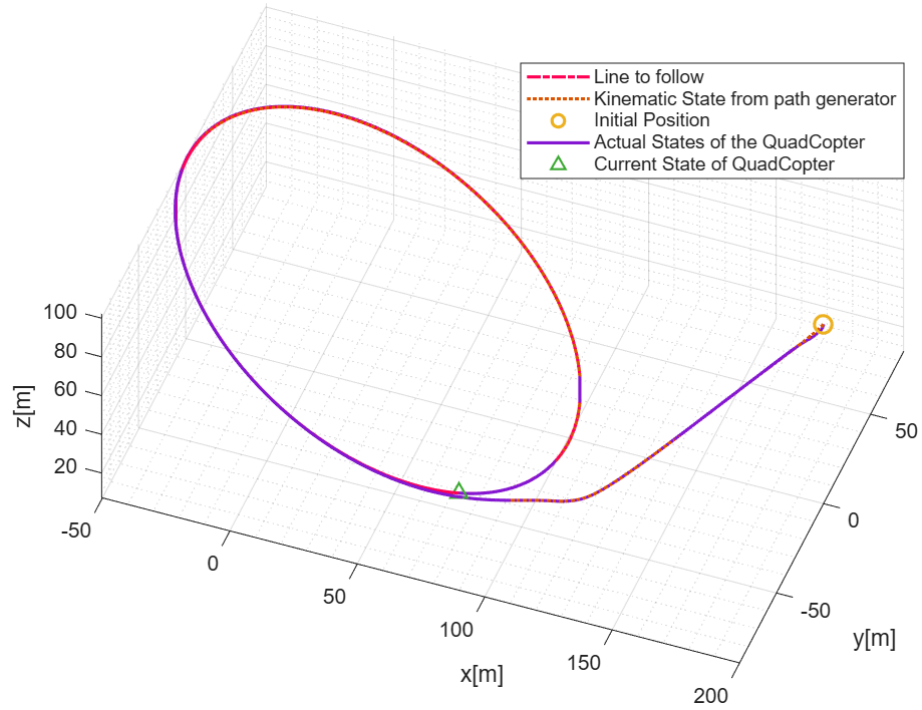
4.3.2.1 Inclined Circle Path Following

For 3D Inclined Circle Path Following using CVF-based guidance with Quadrotor dynamics, the Inclined Circle to follow is given by the curve,

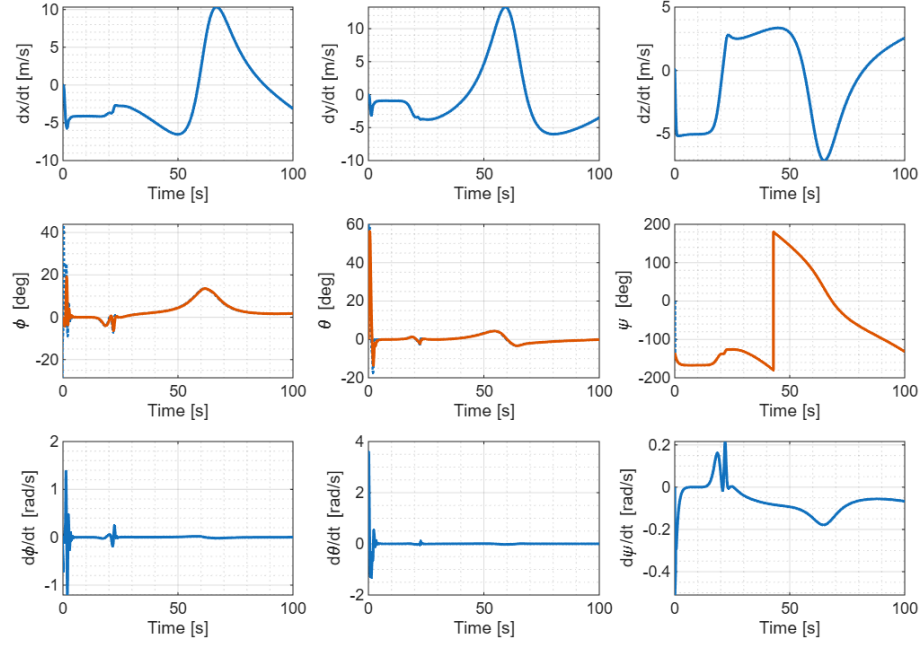
$$f(x, y, z) = \frac{1}{100}((x - 2)^2 + (y + 10)^2 + (z - 15)^2 - 10000) \quad (4.36)$$

$$g(x, y, z) = 3x + y + 5z - 350 \quad (4.37)$$

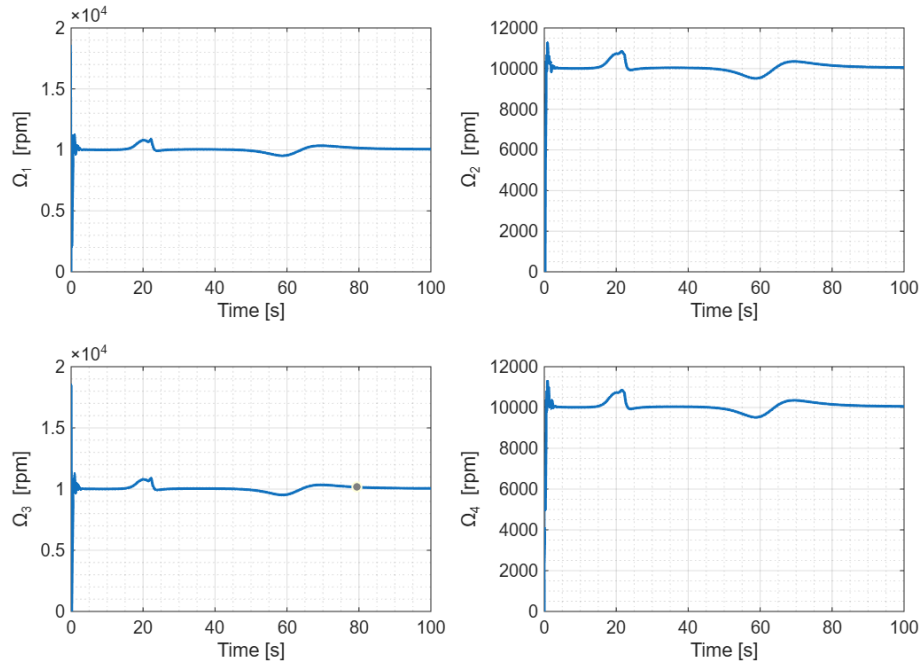
The initial state of the Quadrotor is given as: initial position is $P0(200, 0, 100) m$, initial course angle is $-3\pi/4 rad$, initial climb angle is $-\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 10.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

FIGURE 4.6: 3D Inclined Circle Path Following with Quadrotor Dynamics with CVF-based guidance

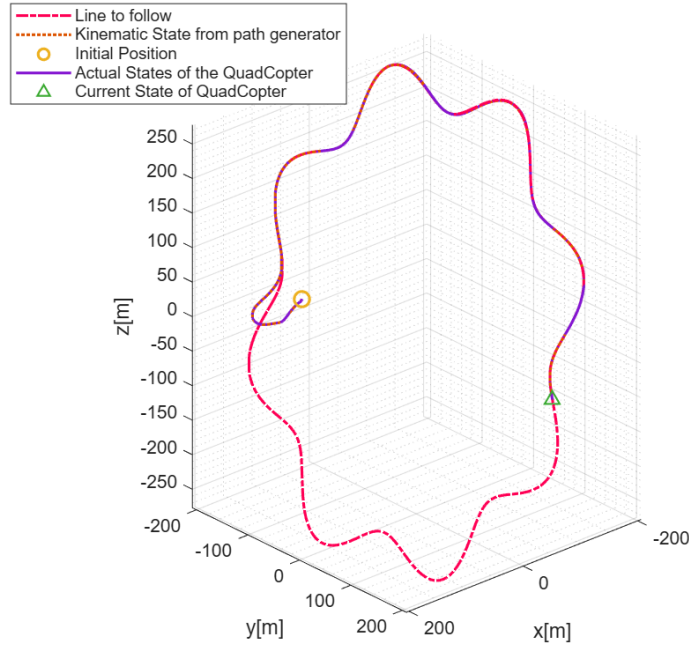
4.3.2.2 Inclined Star Path Following

For 3D Inclined Star Path Following using CVF-based guidance with Quadrotor dynamics, the Inclined Star to follow is given by the curve,

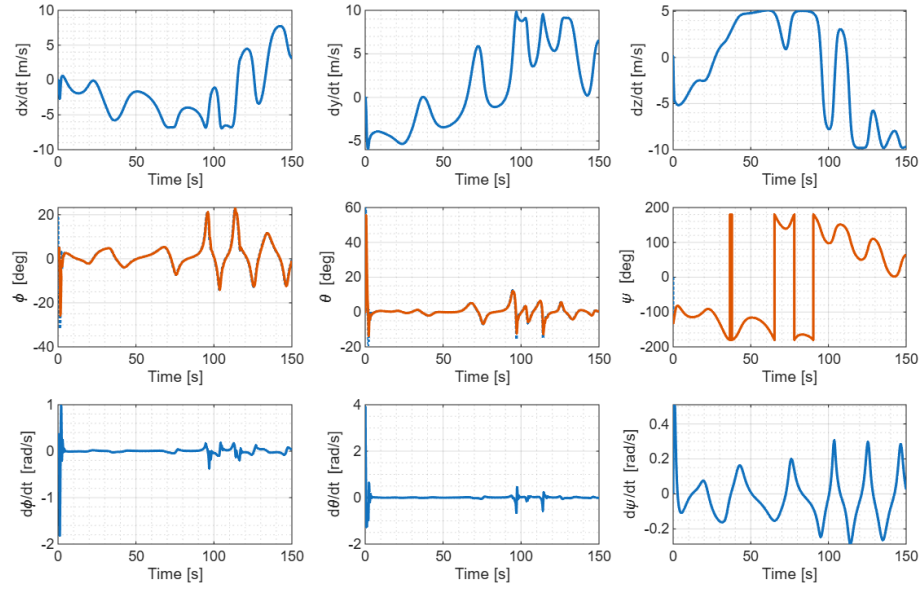
$$f(x, y, z) = \frac{x + y + z}{\sqrt{5x^2 + xy + 5y^2 + 1}} \quad (4.38)$$

$$g(x, y, z) = \frac{1}{(5x^2 + xy + 5y^2)^{\frac{9}{2}} + 1} \left(-(5x^2 + xy + 5y^2)^{\frac{9}{2}} + 300\sqrt{2}(5x^2 + xy + 5y^2)^4 \right. \\ \left. + 45\sqrt{22}(x^2 - y^2)(x^2 + 20xy + y^2)(49x^4 - 20x^3y - 300x^2y^2 - 20xy^3 + 49y^4) \right) \quad (4.39)$$

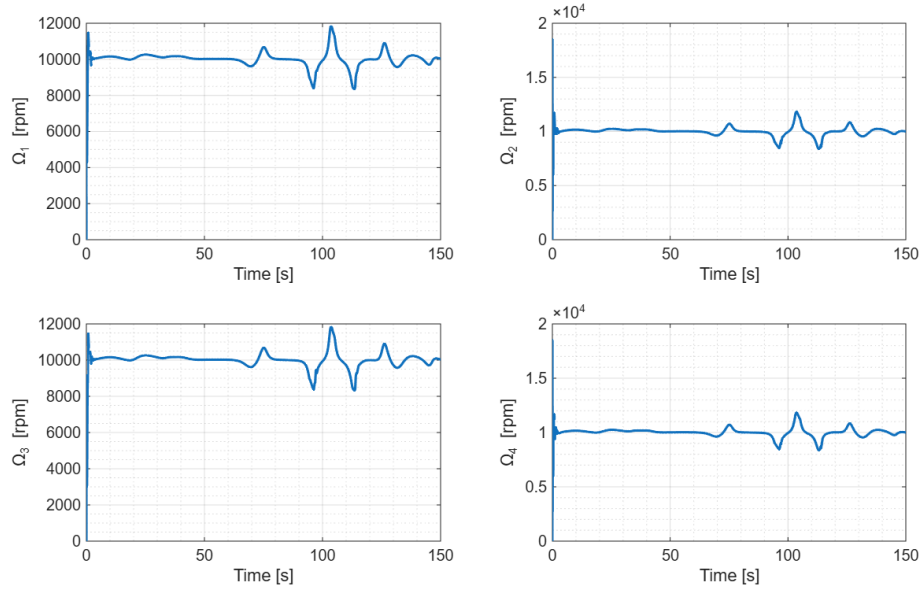
The initial state of the Quadrotor is given as: initial position is $P0(200, 0, 100) m$, initial course angle is $-3\pi/4 rad$, initial climb angle is $-\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 5.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

FIGURE 4.7: 3D Inclined Star Path Following with Quadrotor Dynamics with CVF-based guidance

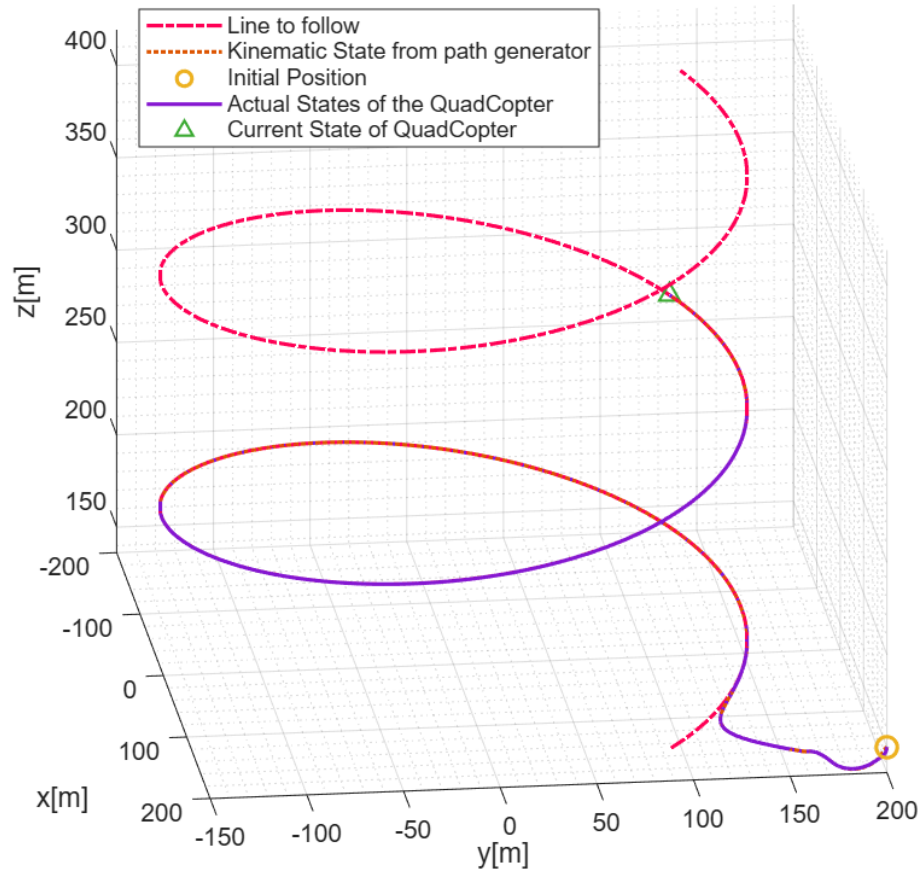
4.3.2.3 Helical Path Following

For 3D Helical Path Following using CVF-based guidance with Quadrotor dynamics, the Helical path to follow is given by the curve,

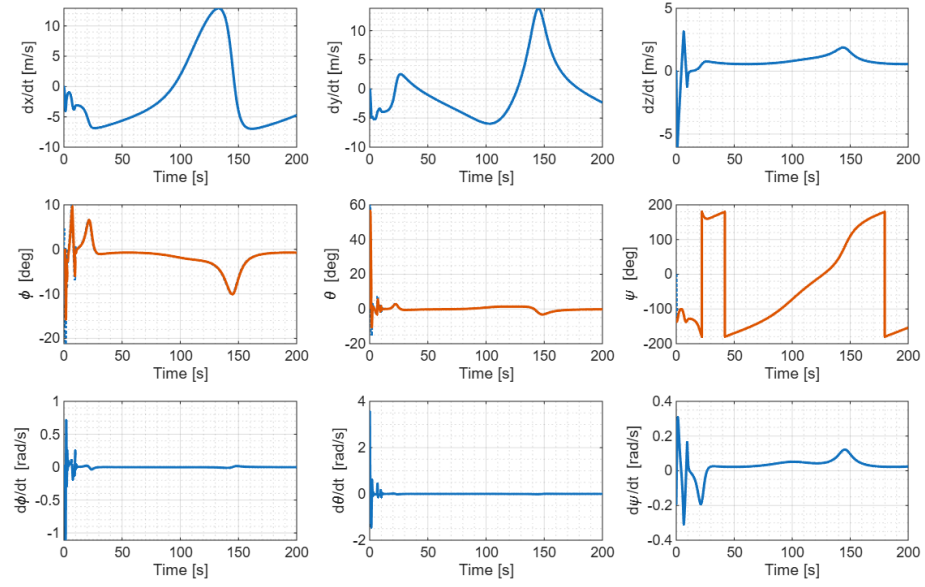
$$f(x, y, z) = x - 200 \cos\left(\frac{z}{20}\right) \quad (4.40)$$

$$g(x, y, z) = y - 150 \sin\left(\frac{z}{20}\right) \quad (4.41)$$

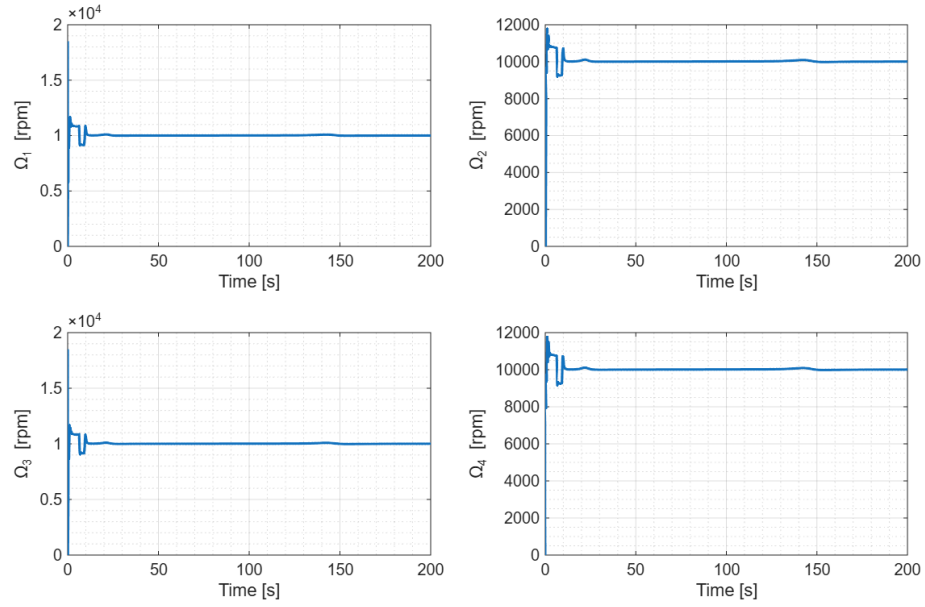
The initial state of the Quadrotor is given as: initial position is $P_0(200, 200, 150) m$, initial course angle is $-3\pi/4 rad$, initial climb angle is $-\pi/4 rad$. The Combined Vector Field design parameter (κ_0) is 8.



(a) Path Following Trajectory



(b) Different parameters Variations with time



(c) Variation in the Motor rotational speed with time

FIGURE 4.8: 3D Helical Path Following with Quadrotor Dynamics with CVF-based guidance

Chapter 5

Conclusion

This thesis explored the use of Combined Vector Fields (CVF) as a guidance strategy for unmanned aerial vehicles, with a particular focus on the motion of Quadrotors. The core idea behind the work was to design a method that guides a vehicle toward a desired path and then helps it follow that path smoothly. The CVF method constructs a direction field in space that naturally "pulls" the vehicle toward the curve and, once near it, encourages it to move along it in a steady and predictable way. Because the method is based on geometry alone, it remains flexible and applicable to a wide variety of two-dimensional and three-dimensional path shapes.

To test this approach in a realistic setting, the guidance algorithm was combined with a quadrotor dynamic model. The lower-level motion stabilization was managed using sliding mode control, providing robustness against minor disturbances and modeling uncertainties. The guidance layer, built on CVFs, generated the desired course and climb commands, which the quadrotor then attempted to follow. The simulations conducted as part of this work showed encouraging results: the quadrotor was able to converge to and follow complex curves such as circles, stars, inclined loops, and spatial helices. The transitions from approaching the curve to tracking along it remained smooth, and the control signals generated were stable and realistic for practical implementation.

However, the study also revealed some challenges. One of the most noticeable limitations was that the CVF method is sensitive to how the path equations are scaled, especially in three dimensions, where curves are represented as the intersection of two surfaces. If one surface dominates the other in magnitude, the vector field may pull the vehicle incorrectly, leading to deviation from the curve. Another limitation is that CVF, in its current form, is designed purely for path following, not time-based trajectory tracking. This means it does not directly control how fast the vehicle moves along the path or ensure that it reaches particular points at specific times. Additionally, all results here are based on simulation. Real-world flight introduces factors such as sensor inaccuracy, actuator saturation, aerodynamic uncertainties, and delays—none of which are fully captured in simulation.

These observations suggest several useful directions for future work. First, an automatic method for scaling or normalizing the implicit path equations would make the CVF approach more reliable across different curve shapes. Second, incorporating timing information into the guidance layer would allow the method to handle full trajectory tracking rather than just geometric path following. Third, obstacle avoidance could be integrated into the vector field framework, enabling the drone to adapt to changing environments while still maintaining the original desired path. Finally, moving from simulation to hardware-in-the-loop testing and actual flight experiments is an essential next step. Such experiments will demonstrate the controller's performance in practical conditions and help refine the design further.

In conclusion, the Combined Vector Field method examined in this thesis provides a clear and intuitive approach to guiding UAVs along complex paths. Its geometric basis makes it both adaptable and conceptually elegant, and the simulation results suggest strong potential for real-world applications. While there are meaningful challenges that remain—particularly in scaling, timing, and experimental validation—the path forward is clear. With continued refinement, the CVF approach can grow into a robust and versatile guidance solution for autonomous aerial systems.

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